

Lecture Notes on Laplace Transform (Lec-13)

1. Definition of Laplace Transform

Let $F(t)$ be a function defined for all $t \geq 0$. Then the Laplace transform of $F(t)$, denoted by $\mathcal{L}\{F(t)\}$ or $f(s)$, is defined as:

$$\mathcal{L}\{F(t)\} = f(s) = \int_0^{\infty} e^{-st} F(t) dt$$

provided that the integral exists.

- **Parameter s :** A parameter which may be real or complex ($s = \alpha + i\omega$).
- **Operator $\mathcal{L}$:** Called the Laplace transform operator.
- **Linearity Property:** The operator $\mathcal{L}$ is linear:

$$\mathcal{L}\{c_1 F_1(t) + c_2 F_2(t)\} = c_1 \mathcal{L}\{F_1(t)\} + c_2 \mathcal{L}\{F_2(t)\}$$

2. Conditions for Existence of Laplace Transform

$\mathcal{L}\{F(t)\}$ exists if:

1. $F(t)$ is piecewise (sectionally) continuous in every finite interval $0 \leq t \leq T$.
2. $F(t)$ is of exponential order as $t \rightarrow \infty$, i.e., $\lim_{t \rightarrow \infty} e^{-\alpha t} F(t) = C$ (where C is a finite number).

Example:

For $F(t) = t^2$:

- $F(t) = t^2$ is a polynomial function, so it is continuous everywhere.
- Checking exponential order:

$$\lim_{t \rightarrow \infty} \frac{t^2}{e^{\alpha t}} = \lim_{t \rightarrow \infty} \frac{2t}{\alpha e^{\alpha t}} = \lim_{t \rightarrow \infty} \frac{2}{\alpha^2 e^{\alpha t}} = 0$$

Thus, $F(t) = t^2$ is of exponential order.

3. Concept Mapping

- **Time Domain (t):** Differential Equations
- **Complex Frequency / s -Domain ($s = \alpha + i\omega$):** Algebraic Equations
 - α : Growth / Decay factor
 - ω : Angular frequency

4. Inverse Laplace Transform

If $f(s)$ is the Laplace transform of a function $F(t)$, i.e., $\mathcal{L}\{F(t)\} = f(s)$, then $F(t)$ is called the inverse Laplace transform of $f(s)$ and is written as:

$$F(t) = \mathcal{L}^{-1}\{f(s)\}$$

where $\mathcal{L}^{-1}$ is called the Inverse Laplace Transform operator. It is also a linear operator.

5. Table of Standard Laplace Transforms & Inverse Transforms

F(t)	$\mathcal{L}\{F(t)\}=f(s)$	Inverse Laplace Transform $\mathcal{L}^{-1}\{f(s)\}=F(t)$
1	$\frac{1}{s}$	$\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} = 1$
t	$\frac{1}{s^2}$	$\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} = t$
t^n ($n = 0, 1, 2, \dots$)	$\frac{n!}{s^{n+1}}$	$\mathcal{L}^{-1}\left\{\frac{1}{s^{n+1}}\right\} = \frac{t^n}{n!}$
t^n ($n \notin \mathbb{Z}_+$)	$\frac{\Gamma(n+1)}{s^{n+1}}$	$\mathcal{L}^{-1}\left\{\frac{1}{s^{n+1}}\right\} = \frac{t^n}{\Gamma(n+1)}$
e^{at}	$\frac{1}{s-a}$	$\mathcal{L}^{-1}\left\{\frac{1}{s-a}\right\} = e^{at}$
$\sin(at)$	$\frac{a}{s^2 + a^2}$	$\mathcal{L}^{-1}\left\{\frac{1}{s^2 + a^2}\right\} = \frac{\sin(at)}{a}$
$\cos(at)$	$\frac{s}{s^2 + a^2}$	$\mathcal{L}^{-1}\left\{\frac{s}{s^2 + a^2}\right\} = \cos(at)$
$\sinh(at)$	$\frac{a}{s^2 - a^2}$	$\mathcal{L}^{-1}\left\{\frac{1}{s^2 - a^2}\right\} = \frac{\sinh(at)}{a}$
$\cosh(at)$	$\frac{s}{s^2 - a^2}$	$\mathcal{L}^{-1}\left\{\frac{s}{s^2 - a^2}\right\} = \cosh(at)$

6. Derivation Proofs Shown in class

Derivation 1: $\mathcal{L}\{1\} = \frac{1}{s}$

$$\mathcal{L}\{1\} = \int_0^{\infty} e^{-s} (1) dt = \left[\frac{e^{-s}}{-s} \right]_0^{\infty} = 0 - \left(-\frac{1}{s} \right) = \frac{1}{s} \quad (s > 0)$$

Derivation 2: $\mathcal{L}\{t^2\} = \frac{2}{s^3}$ using Integration by Parts

$$\mathcal{L}\{t^2\} = \int_0^{\infty} e^{-st} t^2 dt$$

Using Integration by Parts ($\int u dv = uv - \int v du$):

$$\begin{aligned} \mathcal{L}\{t^2\} &= \left[t^2 \frac{e^{-st}}{-s} \right]_0^{\infty} - \int_0^{\infty} 2t \left(\frac{e^{-st}}{-s} \right) dt \\ &= 0 + \frac{2}{s} \int_0^{\infty} t e^{-st} dt \end{aligned}$$

Applying Integration by Parts again to $\int_0^{\infty} t e^{-st} dt$:

$$\begin{aligned} &= \frac{2}{s} \left[\left[t \frac{e^{-st}}{-s} \right]_0^{\infty} - \int_0^{\infty} 1 \cdot \frac{e^{-st}}{-s} dt \right] \\ &= \frac{2}{s} \left[0 + \frac{1}{s} \left[\frac{e^{-s}}{-s} \right]_0^{\infty} \right] \\ &= \frac{2}{s} \cdot \frac{1}{s} \left(\frac{1}{s} \right) = \frac{2}{s^3} \end{aligned}$$

HOME WORK

The general definition of the Laplace transform is:

$$\mathcal{L}\{F(t)\} = \int_0^{\infty} e^{-s} F(t) dt \quad \text{for } s > 0$$

1. Proof for $F(t) = 1$

$$\begin{aligned}\mathcal{L}\{1\} &= \int_0^{\infty} e^{-st} (1) dt \\ &= \left[\frac{e^{-s}}{-s} \right]_0^{\infty} \\ &= \lim_{t \rightarrow \infty} \left(\frac{e^{-s}}{-s} \right) - \left(\frac{e^0}{-s} \right) \\ &= 0 - \left(-\frac{1}{s} \right) = \frac{1}{s}\end{aligned}$$

2. Proof for $F(t) = t^n$

Case A: For positive integer $n \in \mathbb{Z}_+$

$$\mathcal{L}\{t^n\} = \int_0^{\infty} e^{-s} t^n dt$$

Using substitution, let $u = st \Rightarrow t = \frac{u}{s}$ and $dt = \frac{du}{s}$.

- When $t = 0 \Rightarrow u = 0$
- When $t \rightarrow \infty \Rightarrow u \rightarrow \infty$

Substitute these into the integral:

$$\begin{aligned}\mathcal{L}\{t^n\} &= \int_0^\infty e^{-u} \left(\frac{u}{s}\right)^n \left(\frac{du}{s}\right) \\ &= \frac{1}{s^{n+1}} \int_0^\infty e^{-u} u^n du\end{aligned}$$

By the definition of the Gamma function $\Gamma(z) = \int_0^\infty e^{-u} u^{z-1} du$, we have:

$$\int_0^\infty e^{-u} u^n du = \Gamma(n+1)$$

Since n is an integer, $\Gamma(n+1) = n!$.

$$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}$$

Case B: For non-integer $n \notin \mathbb{Z}_+$

Using the same Gamma function integral above:

$$\mathcal{L}\{t^n\} = \frac{\Gamma(n+1)}{s^{n+1}}$$

3. Proof for $F(t) = e^{at}$

$$\begin{aligned}\mathcal{L}\{e^{at}\} &= \int_0^\infty e^{-st} e^{at} dt \\ &= \int_0^\infty e^{-(s-a)t} dt \\ &= \left[\frac{e^{-(s-a)t}}{-(s-a)} \right]_0^\infty \quad \text{for } s > a \\ &= 0 - \left(-\frac{1}{s-a} \right) = \frac{1}{s-a}\end{aligned}$$

4. Proof for $F(t) = \sin(at)$ using Integration by Parts

By definition:

$$\mathcal{L}\{\sin(at)\} = \int_0^{\infty} e^{-st} \sin(at) dt$$

Using the integration by parts formula $\int u dv = uv - \int v du$:

- Let $u = \sin(at) \Rightarrow du = a \cos(at) dt$
- Let $dv = e^{-st} dt \Rightarrow v = -\frac{1}{s} e^{-s}$

$$I = \left[-\frac{1}{s} e^{-st} \sin(at) \right]_0^{\infty} - \int_0^{\infty} \left(-\frac{1}{s} e^{-st} \right) (a \cos(at)) dt$$

The boundary term vanishes because $e^{-\infty} = 0$ and $\sin(0) = 0$:

$$I = 0 + \frac{a}{s} \int_0^{\infty} e^{-st} \cos(at) dt$$

Apply integration by parts a second time to the integral $\int_0^{\infty} e^{-st} \cos(at) dt$:

- Let $u = \cos(at) \Rightarrow du = -a \sin(at) dt$
- Let $dv = e^{-st} dt \Rightarrow v = -\frac{1}{s} e^{-st}$

$$I = \frac{a}{s} \left(\left[-\frac{1}{s} e^{-st} \cos(at) \right]_0^{\infty} - \int_0^{\infty} \left(-\frac{1}{s} e^{-st} \right) (-a \sin(at)) dt \right)$$

Evaluating the boundary term:

$$\left[-\frac{1}{s} e^{-st} \cos(at) \right]_0^{\infty} = 0 - \left(-\frac{1}{s} \cdot 1 \cdot 1 \right) = \frac{1}{s}$$

Substitute this back into the expression:

$$I = \frac{a}{s} \left(\frac{1}{s} - \frac{a}{s} \int_0^{\infty} e^{-st} \sin(at) dt \right)$$

Notice that the integral on the right-hand side is original integral I :

$$I = \frac{a}{s^2} - \frac{a^2}{s^2} I$$

Rearranging terms to solve for I :

$$\begin{aligned} I + \frac{a^2}{s^2} I &= \frac{a}{s^2} \\ I \left(1 + \frac{a^2}{s^2} \right) &= \frac{a}{s^2} \\ I \left(\frac{s^2 + a^2}{s^2} \right) &= \frac{a}{s^2} \end{aligned}$$

Cancel s^2 from both sides:

$$\mathcal{L}\{\sin(at)\} = \frac{a}{s^2 + a^2}$$

Proof for $F(t) = \cos(at)$ using Integration by Parts

By definition:

$$\mathcal{L}\{\cos(at)\} = \int_0^{\infty} e^{-st} \cos(at) dt$$

Using integration by parts:

- Let $u = \cos(at) \Rightarrow du = -a \sin(at) dt$
- Let $dv = e^{-st} dt \Rightarrow v = -\frac{1}{s} e^{-st}$

$$J = \left[-\frac{1}{s} e^{-st} \cos(at) \right]_0^{\infty} - \int_0^{\infty} \left(-\frac{1}{s} e^{-st} \right) (-a \sin(at)) dt$$

Evaluating boundary limits gives $\frac{1}{s}$:

$$J = \frac{1}{s} - \frac{a}{s} \int_0^{\infty} e^{-st} \sin(at) dt$$

Apply integration by parts a second time on the remaining integral:

- Let $u = \sin(at) \Rightarrow du = a \cos(at) dt$
- Let $dv = e^{-st} dt \Rightarrow v = -\frac{1}{s} e^{-st}$

$$J = \frac{1}{s} - \frac{a}{s} \left(\left[-\frac{1}{s} e^{-st} \sin(at) \right]_0^{\infty} - \int_0^{\infty} \left(-\frac{1}{s} e^{-st} \right) (a \cos(at)) dt \right)$$

The boundary term evaluates to 0:

$$J = \frac{1}{s} - \frac{a}{s} \left(0 + \frac{a}{s} \int_0^{\infty} e^{-st} \cos(at) dt \right)$$

Substitute the original integral J :

$$J = \frac{1}{s} - \frac{a^2}{s^2} J$$

Rearranging terms to solve for J :

$$J + \frac{a^2}{s^2} J = \frac{1}{s}$$
$$J \left(\frac{s^2 + a^2}{s^2} \right) = \frac{1}{s}$$

$$J = \frac{1}{s} \cdot \frac{s^2}{s^2 + a^2}$$

$$\mathcal{L}\{\cos(at)\} = \frac{s}{s^2 + a^2}$$

Alternative

4. Proof for $F(t) = \sin(at)$ and $F(t) = \cos(at)$

Using Euler's formula $e^{iat} = \cos(at) + i \sin(at)$:

$$\mathcal{L}\{e^{iat}\} = \mathcal{L}\{\cos(at)\} + i\mathcal{L}\{\sin(at)\}$$

Using the exponential rule derived above with $a \rightarrow ia$:

$$\mathcal{L}\{e^{iat}\} = \frac{1}{s - ia}$$

Multiply the numerator and denominator by the complex conjugate $(s + ia)$:

$$\frac{1}{s - ia} = \frac{s + ia}{(s - ia)(s + ia)} = \frac{s + ia}{s^2 + a^2}$$

$$= \frac{s}{s^2 + a^2} + i \frac{a}{s^2 + a^2}$$

Equating the real and imaginary parts from both sides:

- **Real part:**

$$\mathcal{L}\{\cos(at)\} = \frac{s}{s^2 + a^2}$$

- **Imaginary part:**

$$\mathcal{L}\{\sin(at)\} = \frac{a}{s^2 + a^2}$$

5.Proof for $F(t) = \sinh(at)$ using Integration by Parts

By definition:

$$\mathcal{L}\{\sinh(at)\} = \int_0^{\infty} e^{-s} \sinh(at) dt$$

Using integration by parts ($\int u dv = uv - \int v du$):

- Let $u = \sinh(at) \Rightarrow du = a \cosh(at) dt$
- Let $dv = e^{-s} dt \Rightarrow v = -\frac{1}{s} e^{-st}$

$$I = \left[-\frac{1}{s} e^{-st} \sinh(at) \right]_0^{\infty} - \int_0^{\infty} \left(-\frac{1}{s} e^{-st} \right) (a \cosh(at)) dt$$

Evaluating boundary limits (since $\sinh(0) = 0$ and $e^{-st} \rightarrow 0$ as $t \rightarrow \infty$ for $s > a$):

$$I = 0 + \frac{a}{s} \int_0^{\infty} e^{-st} \cosh(at) dt$$

Apply integration by parts a second time to $\int_0^{\infty} e^{-st} \cosh(at) dt$:

- Let $u = \cosh(at) \Rightarrow du = a \sinh(at) dt$
- Let $dv = e^{-st} dt \Rightarrow v = -\frac{1}{s} e^{-st}$

$$I = \frac{a}{s} \left(\left[-\frac{1}{s} e^{-st} \cosh(at) \right]_0^{\infty} - \int_0^{\infty} \left(-\frac{1}{s} e^{-st} \right) (a \sinh(at)) dt \right)$$

Evaluating the boundary term (since $\cosh(0) = 1$ and $e^{-\infty} = 0$):

$$\left[-\frac{1}{s} e^{-st} \cosh(at) \right]_0^{\infty} = 0 - \left(-\frac{1}{s} \cdot 1 \right) = \frac{1}{s}$$

Substitute this back into the expression:

$$I = \frac{a}{s} \left(\frac{1}{s} + \frac{a}{s} \int_0^{\infty} e^{-st} \sinh(at) dt \right)$$

Substitute the original integral I :

$$I = \frac{a}{s^2} + \frac{a^2}{s^2} I$$

Rearranging terms to solve for I :

$$I - \frac{a^2}{s^2} I = \frac{a}{s^2}$$

$$I\left(1 - \frac{a^2}{s^2}\right) = \frac{a}{s^2}$$

$$I\left(\frac{s^2 - a^2}{s^2}\right) = \frac{a}{s^2}$$

Cancel s^2 from both sides:

$$\mathcal{L}\{\sinh(at)\} = \frac{a}{s^2 - a^2} \quad (s > |a|)$$

Proof for $F(t) = \cosh(at)$ using Integration by Parts

By definition:

$$\mathcal{L}\{\cosh(at)\} = \int_0^\infty e^{-s} \cosh(at) dt$$

Using integration by parts:

- Let $u = \cosh(at) \Rightarrow du = a \sinh(at) dt$
- Let $dv = e^{-st} dt \Rightarrow v = -\frac{1}{s} e^{-st}$

$$J = \left[-\frac{1}{s} e^{-st} \cosh(at) \right]_0^\infty - \int_0^\infty \left(-\frac{1}{s} e^{-st} \right) (a \sinh(at)) dt$$

Evaluating boundary limits gives $\frac{1}{s}$:

$$J = \frac{1}{s} + \frac{a}{s} \int_0^\infty e^{-st} \sinh(at) dt$$

Apply integration by parts a second time on the remaining integral:

- Let $u = \sinh(at) \Rightarrow du = a \cosh(at) dt$
- Let $dv = e^{-st} dt \Rightarrow v = -\frac{1}{s} e^{-st}$

$$J = \frac{1}{s} + \frac{a}{s} \left(\left[-\frac{1}{s} e^{-st} \sinh(at) \right]_0^\infty - \int_0^\infty \left(-\frac{1}{s} e^{-st} \right) (a \cosh(at)) dt \right)$$

The boundary term evaluates to 0:

$$J = \frac{1}{s} + \frac{a}{s} \left(0 + \frac{a}{s} \int_0^\infty e^{-st} \cosh(at) dt \right)$$

Substitute the original integral J :

$$J = \frac{1}{s} + \frac{a^2}{s^2} J$$

Rearranging terms to solve for J :

$$\begin{aligned}
 J - \frac{a^2}{s^2}J &= \frac{1}{s} \\
 J\left(\frac{s^2 - a^2}{s^2}\right) &= \frac{1}{s} \\
 J &= \frac{1}{s} \cdot \frac{s^2}{s^2 - a^2} \\
 \mathcal{L}\{\cosh(at)\} &= \frac{\mathbf{s}}{\mathbf{s}^2 - \mathbf{a}^2} \quad (s > |a|)
 \end{aligned}$$

Alternative

5. Proof for $F(t) = \sinh(at)$ and $F(t) = \cosh(at)$

For $\sinh(at) = \frac{e^{at} - e^{-at}}{2}$:

$$\begin{aligned}
 \mathcal{L}\{\sinh(at)\} &= \frac{1}{2}[\mathcal{L}\{e^{at}\} - \mathcal{L}\{e^{-at}\}] \\
 &= \frac{1}{2}\left[\frac{1}{s-a} - \frac{1}{s+a}\right] \\
 &= \frac{1}{2}\left[\frac{(s+a) - (s-a)}{s^2 - a^2}\right] \\
 &= \frac{1}{2}\left[\frac{2a}{s^2 - a^2}\right] = \frac{\mathbf{a}}{\mathbf{s}^2 - \mathbf{a}^2}
 \end{aligned}$$

For $\cosh(at) = \frac{e^{at} + e^{-at}}{2}$:

$$\begin{aligned}
 \mathcal{L}\{\cosh(at)\} &= \frac{1}{2}[\mathcal{L}\{e^{at}\} + \mathcal{L}\{e^{-at}\}] \\
 &= \frac{1}{2}\left[\frac{1}{s-a} + \frac{1}{s+a}\right] \\
 &= \frac{1}{2}\left[\frac{(s+a) + (s-a)}{s^2 - a^2}\right] \\
 &= \frac{1}{2}\left[\frac{2s}{s^2 - a^2}\right] = \frac{\mathbf{s}}{\mathbf{s}^2 - \mathbf{a}^2}
 \end{aligned}$$

Method 1: Direct Definition using Integration by Parts

By the definition of the Laplace transform:

$$\mathcal{L}\{t\} = \int_0^{\infty} e^{-st} t \, dt$$

Using Integration by Parts ($\int u \, dv = uv - \int v \, du$):

- Let $u = t \Rightarrow du = dt$
- Let $dv = e^{-st} \, dt \Rightarrow v = \frac{e^{-st}}{-s}$

$$\mathcal{L}\{t\} = \left[t \cdot \left(\frac{e^{-st}}{-s} \right) \right]_0^{\infty} - \int_0^{\infty} \left(\frac{e^{-st}}{-s} \right) dt$$

Evaluating the first term at the limits for $s > 0$:

- As $t \rightarrow \infty$, $\lim_{t \rightarrow \infty} \frac{-t}{se^{st}} = 0$ (by L'Hôpital's Rule).
- At $t = 0$, $0 \cdot \left(\frac{1}{-s} \right) = 0$.

So the boundary term vanishes, leaving:

$$\mathcal{L}\{t\} = 0 + \frac{1}{s} \int_0^{\infty} e^{-st} \, dt$$

Now, integrate e^{-st} :

$$\begin{aligned} \mathcal{L}\{t\} &= \frac{1}{s} \left[\frac{e^{-st}}{-s} \right]_0^{\infty} \\ &= \frac{1}{s} \left(0 - \left(-\frac{1}{s} \right) \right) \\ &= \frac{1}{s^2} \quad (s > 0) \end{aligned}$$

Method 2: Using the General Formula for t^n

For any non-negative integer n , the general Laplace transform formula is:

$$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}$$

Setting $n = 1$:

$$\mathcal{L}\{t^1\} = \frac{1!}{s^{1+1}} = \frac{\mathbf{1}}{\mathbf{s}^2}$$

Lecture- 14

1. Properties of Laplace & Inverse Laplace Transforms

(1) Linearity

$$\begin{aligned}\mathcal{L}\{c_1 F_1(t) + c_2 F_2(t)\} &= c_1 f_1(s) + c_2 f_2(s) \\ \mathcal{L}^{-1}\{c_1 f_1(s) + c_2 f_2(s)\} &= c_1 \mathcal{L}^{-1}\{f_1(s)\} + c_2 \mathcal{L}^{-1}\{f_2(s)\}\end{aligned}$$

(2) First Translation (Shifting) Theorem

$$\begin{aligned}\mathcal{L}\{e^{at} F(t)\} &= f(s - a) \\ \mathcal{L}^{-1}\{f(s - a)\} &= e^{at} F(t)\end{aligned}$$

(3) Second Translation (Shifting) Theorem

If $G(t) = \begin{cases} F(t - a), & t > a \\ 0, & t < a \end{cases} = u(t - a)F(t - a)$, then:

$$\begin{aligned}\mathcal{L}\{u(t - a)F(t - a)\} &= e^{-as} f(s) \\ \mathcal{L}^{-1}\{e^{-as} f(s)\} &= u(t - a)F(t - a)\end{aligned}$$

(4) Change of Scale Property

$$\begin{aligned}\mathcal{L}\{F(at)\} &= \frac{1}{a} f\left(\frac{s}{a}\right) \\ \mathcal{L}^{-1}\{f(ks)\} &= \frac{1}{k} F\left(\frac{t}{k}\right)\end{aligned}$$

(5) Laplace Transform of Derivatives

$$\mathcal{L}\{F^{(n)}(t)\} = s^n f(s) - s^{n-1} F(0) - s^{n-2} F'(0) - \dots - F^{(n-1)}(0)$$

- **First Derivative:** $\mathcal{L}\{F'(t)\} = sf(s) - F(0)$
- **Second Derivative:** $\mathcal{L}\{F''(t)\} = s^2 f(s) - sF(0) - F'(0)$
- **Third Derivative:** $\mathcal{L}\{F'''(t)\} = s^3 f(s) - s^2 F(0) - sF'(0) - F''(0)$

(6) Laplace Transform of Integrals

$$\mathcal{L}\left\{\int_0^t F(u) du\right\} = \frac{f(s)}{s}$$

$$\mathcal{L}^{-1}\left\{\frac{f(s)}{s}\right\} = \int_0^t F(u) du$$

(7) Multiplication by t^n (Derivative of $f(s)$)

$$\mathcal{L}\{t^n F(t)\} = (-1)^n \frac{d^n}{ds^n} [f(s)]$$

$$\mathcal{L}^{-1}\left\{\frac{d^n}{ds^n} [f(s)]\right\} = (-1)^n t^n F(t)$$

(8) Division by t (Integral of $f(s)$)

$$\mathcal{L}\left\{\frac{F(t)}{t}\right\} = \int_s^\infty f(u) du$$

$$\mathcal{L}^{-1}\left\{\int_s^\infty f(u) du\right\} = \frac{F(t)}{t}$$

(9) Derivative of Delta/Impulse Properties

$$\mathcal{L}^{-1}\{s^k\} = 0, \quad \text{for } k = 0, 1, 2, \dots \text{ (in distributional sense)}$$

(10) Periodic Functions

If $F(t)$ is periodic with period T , so that $F(t + T) = F(t)$, then:

$$\mathcal{L}\{F(t)\} = \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} F(t) dt$$

Unit Step Function Definition

The Heaviside Unit Step Function $u(t - a)$ is defined as:

$$u(t - a) = \begin{cases} 0, & t < a \\ 1, & t \geq a \end{cases}$$

Examples Shown in class

Example 1: Evaluate $\mathcal{L}\{e^{-7t} + 2t^2 + 5 \cos(6t) - 7 \sin(4t)\}$

$$\mathcal{L}\{e^{-7t} + 2t^2 + 5 \cos(6t) - 7 \sin(4t)\} = \mathcal{L}\{e^{-7t}\} + 2\mathcal{L}\{t^2\} + 5\mathcal{L}\{\cos(6t)\} - 7\mathcal{L}\{\sin(4t)\}$$

Using standard formulas:

- $\mathcal{L}\{e^{-7t}\} = \frac{1}{s+7}$
- $\mathcal{L}\{t^2\} = \frac{2!}{s^3} = \frac{2}{s^3}$
- $\mathcal{L}\{\cos(6t)\} = \frac{s}{s^2+36}$
- $\mathcal{L}\{\sin(4t)\} = \frac{4}{s^2+16}$

Final Answer:

$$= \frac{1}{s+7} + \frac{4}{s^3} + \frac{5s}{s^2+36} - \frac{28}{s^2+16}$$

Example 2: Evaluate $\mathcal{L}\{e^{-2t} \sin(4t)\}$

Let $F(t) = \sin(4t)$.

$$f(s) = \mathcal{L}\{\sin(4t)\} = \frac{4}{s^2+16}$$

By the **1st Translation Theorem**, replace $s \rightarrow s + 2$:

$$\mathcal{L}\{e^{-2t} \sin(4t)\} = \frac{4}{(s+2)^2+16} = \frac{4}{s^2+4s+20}$$

Example 3: Evaluate $\mathcal{L}\{e^{-3t}(3 \cos(6t) - 5 \sin(6t))\}$

$$\mathcal{L}\{e^{-3t}(3 \cos(6t) - 5 \sin(6t))\} = 3\mathcal{L}\{e^{-3t} \cos(6t)\} - 5\mathcal{L}\{e^{-3t} \sin(6t)\}$$

Let $F(t) = 3 \cos(6t) - 5 \sin(6t)$:

$$f(s) = 3 \cdot \frac{s}{s^2+36} - 5 \cdot \frac{6}{s^2+36} = \frac{3s-30}{s^2+36}$$

By applying shifting $s \rightarrow s + 3$:

$$\mathcal{L}\{e^{-3t}(3 \cos(6t) - 5 \sin(6t))\} = \frac{3(s+3) - 30}{(s+3)^2 + 36} = \frac{3s - 21}{s^2 + 6s + 45}$$

Problem 4.7: Evaluate $\mathcal{L}\{F(t)\}$ for Piecewise Function

$$F(t) = \begin{cases} 3, & 0 < t < 2 \\ -1, & 2 \leq t < 4 \\ 0, & t \geq 4 \end{cases}$$

By definition:

$$\begin{aligned} \mathcal{L}\{F(t)\} &= \int_0^{\infty} e^{-st} F(t) dt \\ &= \int_0^2 e^{-st} (3) dt + \int_2^4 e^{-st} (-1) dt + \int_4^{\infty} e^{-s} (0) dt \\ &= 3 \left[\frac{e^{-st}}{-s} \right]_0^2 - 1 \left[\frac{e^{-st}}{-s} \right]_2^4 \\ &= -\frac{3}{s}(e^{-2s} - 1) + \frac{1}{s}(e^{-4s} - e^{-2s}) \\ &= \frac{3 - 4e^{-2s} + e^{-4s}}{s} \end{aligned}$$

Problem 4.17: Show that $\mathcal{L}\{u(t-a)\} = \frac{e^{-as}}{s}$, $s > 0$

Proof:

$$u(t-a) = \begin{cases} 0, & t < a \\ 1, & t \geq a \end{cases}$$

By definition:

$$\begin{aligned} \mathcal{L}\{u(t-a)\} &= \int_0^{\infty} e^{-st} u(t-a) dt = \int_a^{\infty} e^{-st} (1) dt \\ &= \left[\frac{e^{-st}}{-s} \right]_a^{\infty} = 0 - \left(\frac{e^{-as}}{-s} \right) = \frac{e^{-as}}{s} \end{aligned}$$

Problem 4.18: Express in terms of Unit Step Function and find $\mathcal{L}\{f(t)\}$

$$f(t) = \begin{cases} 8, & 0 \leq t \leq 2 \\ 6, & t > 2 \end{cases}$$

Expressing $f(t)$ using $u(t - 2)$:

$$f(t) = 8 - 2u(t - 2)$$

Taking Laplace Transform:

$$\mathcal{L}\{f(t)\} = 8\mathcal{L}\{1\} - 2\mathcal{L}\{u(t - 2)\}$$

$$\mathcal{L}\{f(t)\} = \frac{8}{s} - \frac{2e^{-2s}}{s}$$

Problem 4.8: Evaluate $\mathcal{L}\{\sin(t) \cos(t)\}$

Using trigonometric identity $\sin(t) \cos(t) = \frac{1}{2} \sin(2t)$:

$$\begin{aligned} \mathcal{L}\{\sin(t) \cos(t)\} &= \mathcal{L}\left\{\frac{1}{2} \sin(2t)\right\} \\ &= \frac{1}{2} \cdot \frac{2}{s^2 + 4} = \frac{1}{s^2 + 4} \end{aligned}$$

Assigned Homework Problems

- HW: 4.3, 4.5, 4.6, 4.48, 4.49, 4.60, 4.63, 4.80

HOMEWORK SOLUTION

4.3. Prove that $\mathcal{L}$ is a linear operator.

Proof:

We must show that if c_1, c_2 are any constants and $f_1(t), f_2(t)$ are any functions whose Laplace transforms exist, then:

$$\mathcal{L}\{c_1 f_1(t) + c_2 f_2(t)\} = c_1 \mathcal{L}\{f_1(t)\} + c_2 \mathcal{L}\{f_2(t)\}$$

Using the definition of the Laplace transform:

$$\mathcal{L}\{c_1 f_1(t) + c_2 f_2(t)\} = \int_0^{\infty} e^{-st} \{c_1 f_1(t) + c_2 f_2(t)\} dt$$

Distributing e^{-st} and splitting the integral by linearity of integration:

$$= c_1 \int_0^{\infty} e^{-st} f_1(t) dt + c_2 \int_0^{\infty} e^{-st} f_2(t) dt$$

Substituting back the definition of the Laplace transform:

$$= c_1 \mathcal{L}\{f_1(t)\} + c_2 \mathcal{L}\{f_2(t)\}$$

And so, $\mathcal{L}$ is a linear operator. ■

4.5. Find the Laplace transforms of each of the following:

(a) $3e^{-4t}$

$$\mathcal{L}\{3e^{-4t}\} = 3\mathcal{L}\{e^{-4t}\} = \frac{3}{s - (-4)} = \frac{3}{s + 4}, \quad s > -4$$

(b) $2t^2$

$$\mathcal{L}\{2t^2\} = 2\mathcal{L}\{t^2\} = \frac{2\Gamma(3)}{s^3} = \frac{2 \cdot 2!}{s^3} = \frac{4}{s^3}, \quad s > 0$$

(c) $4 \cos(5t)$

$$\mathcal{L}\{4 \cos(5t)\} = 4\mathcal{L}\{\cos(5t)\} = 4 \cdot \frac{s}{s^2 + 25} = \frac{4s}{s^2 + 25}, \quad s > 0$$

(d) $\sin(\pi t)$

$$\mathcal{L}\{\sin(\pi t)\} = \frac{\pi}{s^2 + \pi^2}, \quad s > 0$$

(e) $-\frac{3}{\sqrt{t}}$

$$\mathcal{L}\left\{-\frac{3}{\sqrt{t}}\right\} = -3\mathcal{L}\{t^{-1/2}\} = -\frac{3\Gamma(1/2)}{s^{1/2}}$$

Since $\Gamma(1/2) = \sqrt{\pi}$:

$$= -\frac{3\sqrt{\pi}}{\sqrt{s}} = -3\sqrt{\frac{\pi}{s}}, \quad s > 0$$

4.6. Find the Laplace transforms of each of the following:

(a) $3t^4 - 2t^{3/2} + 6$

$$\begin{aligned}\mathcal{L}\{3t^4 - 2t^{3/2} + 6\} &= 3\mathcal{L}\{t^4\} - 2\mathcal{L}\{t^{3/2}\} + 6\mathcal{L}\{1\} \\ &= \frac{3\Gamma(5)}{s^5} - \frac{2\Gamma(5/2)}{s^{5/2}} + \frac{6}{s}\end{aligned}$$

Since $\Gamma(5) = 4! = 24$ and $\Gamma(5/2) = \frac{3}{2} \cdot \frac{1}{2} \Gamma(1/2) = \frac{3}{4}\sqrt{\pi}$:

$$= \frac{3 \cdot 24}{s^5} - \frac{2 \cdot \left(\frac{3}{4}\sqrt{\pi}\right)}{s^{5/2}} + \frac{6}{s} = \frac{72}{s^5} - \frac{3\sqrt{\pi}}{2s^{5/2}} + \frac{6}{s}$$

(b) $5 \sin(2t) - 3 \cos(2t)$

$$\begin{aligned}\mathcal{L}\{5 \sin(2t) - 3 \cos(2t)\} &= 5\mathcal{L}\{\sin(2t)\} - 3\mathcal{L}\{\cos(2t)\} \\ &= \frac{5 \cdot 2}{s^2 + 4} - \frac{3 \cdot s}{s^2 + 4} = \frac{10 - 3s}{s^2 + 4}\end{aligned}$$

(c) $3\sqrt[3]{t} + 4e^{2t}$

$$\mathcal{L}\{3\sqrt[3]{t} + 4e^{2t}\} = 3\mathcal{L}\{t^{1/3}\} + 4\mathcal{L}\{e^{2t}\} = \frac{3\Gamma(4/3)}{s^{4/3}} + \frac{4}{s-2}$$

Using $\Gamma(4/3) = \frac{1}{3}\Gamma(1/3)$:

$$= \frac{3 \cdot \frac{1}{3}\Gamma(1/3)}{s^{4/3}} + \frac{4}{s-2} = \frac{\Gamma(1/3)}{s^{4/3}} + \frac{4}{s-2}$$

(d) $\frac{1}{t^2}$

$$\mathcal{L}\left\{\frac{1}{t^2}\right\} = \int_0^{\infty} \frac{e^{-st}}{t^2} dt$$

As $t \rightarrow 0^+$, the integrand $\frac{e^{-st}}{t^2} \sim \frac{1}{t^2}$, and $\int_0^{\epsilon} \frac{1}{t^2} dt = \left[-\frac{1}{t}\right]_0^{\epsilon} = \infty$.

Since this integral **diverges** at the lower limit $t = 0$, the Laplace transform **does not exist**

4.48. Find the Laplace transforms of each of the following:

(a) $4e^{2t/3}$

$$\mathcal{L}\{4e^{2t/3}\} = 4\mathcal{L}\{e^{(2/3)t}\} = 4 \cdot \frac{1}{s - \frac{2}{3}} = 4 \cdot \frac{1}{\frac{3s - 2}{3}} = \frac{12}{3s - 2}$$

(b) $6t - 3$

$$\mathcal{L}\{6t - 3\} = 6\mathcal{L}\{t\} - 3\mathcal{L}\{1\} = 6 \cdot \frac{1}{s^2} - 3 \cdot \frac{1}{s} = \frac{6}{s^2} - \frac{3s}{s^2} = \frac{6 - 3s}{s^2}$$

(c) $(t + 1)^2$

Expand the binomial:

$$(t + 1)^2 = t^2 + 2t + 1$$

Take the Laplace transform term-by-term:

$$\mathcal{L}\{t^2 + 2t + 1\} = \mathcal{L}\{t^2\} + 2\mathcal{L}\{t\} + \mathcal{L}\{1\} = \frac{2}{s^3} + \frac{2}{s^2} + \frac{1}{s} = \frac{s^2 + 2s + 2}{s^3}$$

(d) $2 \sin(3t) + 5 \cos(3t)$

$$\begin{aligned} \mathcal{L}\{2 \sin(3t) + 5 \cos(3t)\} &= 2\mathcal{L}\{\sin(3t)\} + 5\mathcal{L}\{\cos(3t)\} \\ &= 2 \cdot \frac{3}{s^2 + 9} + 5 \cdot \frac{s}{s^2 + 9} = \frac{6 + 5s}{s^2 + 9} = \frac{5s + 6}{s^2 + 9} \end{aligned}$$

(e) $(e^{3t} - e^{-3t})^2$

Expand the squared term:

$$(e^{3t} - e^{-3t})^2 = e^{6t} - 2e^{3t}e^{-3t} + e^{-6t} = e^{6t} - 2 + e^{-6t}$$

Take the Laplace transform:

$$\begin{aligned}
 \mathcal{L}\{e^{6t} - 2 + e^{-6t}\} &= \frac{1}{s-6} - \frac{2}{s} + \frac{1}{s+6} \\
 &= \left(\frac{1}{s-6} + \frac{1}{s+6}\right) - \frac{2}{s} = \frac{2s}{s^2-36} - \frac{2}{s} \\
 &= \frac{2s^2 - 2(s^2-36)}{s(s^2-36)} = \frac{2s^2 - 2s^2 + 72}{s(s^2-36)} = \frac{72}{s(s^2-36)}
 \end{aligned}$$

(f) $\frac{(\sqrt{t}+1)(2-\sqrt{t})}{\sqrt{t}}$

Expand the numerator:

$$(\sqrt{t}+1)(2-\sqrt{t}) = 2\sqrt{t} - t + 2 - \sqrt{t} = \sqrt{t} + 2 - t$$

Divide by $\sqrt{t}$:

$$f(t) = \frac{\sqrt{t} + 2 - t}{\sqrt{t}} = 1 + \frac{2}{\sqrt{t}} - \sqrt{t} = 1 + 2t^{-1/2} - t^{1/2}$$

Take the Laplace transform using $\mathcal{L}\{t^p\} = \frac{\Gamma(p+1)}{s^{p+1}}$:

$$\begin{aligned}
 \mathcal{L}\{f(t)\} &= \mathcal{L}\{1\} + 2\mathcal{L}\{t^{-1/2}\} - \mathcal{L}\{t^{1/2}\} \\
 &= \frac{1}{s} + \frac{2\Gamma(1/2)}{s^{1/2}} - \frac{\Gamma(3/2)}{s^{3/2}}
 \end{aligned}$$

Using $\Gamma(1/2) = \sqrt{\pi}$ and $\Gamma(3/2) = \frac{1}{2}\sqrt{\pi}$:

$$= \frac{1}{s} + \frac{2\sqrt{\pi}}{\sqrt{s}} - \frac{3\sqrt{\pi}}{4s^{5/2}}$$

(g) $8 \sin^2(3t)$

Use the trigonometric identity $\sin^2(\theta) = \frac{1-\cos(2\theta)}{2}$:

$$8 \sin^2(3t) = 8 \left(\frac{1 - \cos(6t)}{2} \right) = 4 - 4 \cos(6t)$$

Take the Laplace transform:

$$\mathcal{L}\{4 - 4 \cos(6t)\} = \frac{4}{s} - \frac{4s}{s^2 + 36} = 4 \left(\frac{(s^2 + 36) - s^2}{s(s^2 + 36)} \right) = \frac{144}{s(s^2 + 36)}$$

(h) $\sin(2t) \cos(2t)$

Use the double-angle identity $\sin(2\theta) = 2 \sin(\theta) \cos(\theta)$:

$$\sin(2t) \cos(2t) = \frac{1}{2} \sin(4t)$$

Take the Laplace transform:

$$\mathcal{L}\left\{\frac{1}{2} \sin(4t)\right\} = \frac{1}{2} \cdot \frac{4}{s^2 + 16} = \frac{2}{s^2 + 16}$$

(i) $(\sqrt[3]{t^2} - 1/\sqrt[3]{t})^2$

Rewrite exponents and expand:

$$\left(\sqrt[3]{t^2} - 1/\sqrt[3]{t}\right)^2 = (t^{2/3} - t^{-1/3})^2 = t^{4/3} - 2t^{1/3} + t^{-2/3}$$

Take the Laplace transform using $\mathcal{L}\{t^p\} = \frac{\Gamma(p+1)}{s^{p+1}}$:

$$\mathcal{L}\{t^{4/3}\} = \frac{\Gamma(7/3)}{s^{7/3}} = \frac{4}{3} \cdot \frac{1}{3} \Gamma(1/3) = \frac{4\Gamma(1/3)}{9s^{7/3}}$$

$$\mathcal{L}\{2t^{1/3}\} = \frac{2\Gamma(4/3)}{s^{4/3}} = \frac{2 \cdot \frac{1}{3} \Gamma(1/3)}{s^{4/3}} = \frac{2\Gamma(1/3)}{3s^{4/3}}$$

$$\mathcal{L}\{t^{-2/3}\} = \frac{\Gamma(1/3)}{s^{1/3}} \quad \text{or expressing directly as } \frac{\Gamma(2/3)}{s^{2/3}}$$

Combining all terms:

$$= \frac{4\Gamma(1/3)}{9s^{7/3}} - \frac{2\Gamma(1/3)}{3s^{4/3}} + \frac{\Gamma(2/3)}{s^{2/3}}$$

(j) $5 \sinh(2t) - 5 \cosh(2t)$

Using definitions $\sinh(2t) = \frac{e^{2t} - e^{-2t}}{2}$ and $\cosh(2t) = \frac{e^{2t} + e^{-2t}}{2}$:

$$5 \sinh(2t) - 5 \cosh(2t) = 5 \left(\frac{e^{2t} - e^{-2t}}{2} - \frac{e^{2t} + e^{-2t}}{2} \right) = 5(-e^{-2t}) = -5e^{-2t}$$

Take the Laplace transform:

$$\mathcal{L}\{-5e^{-2t}\} = -5 \cdot \frac{1}{s - (-2)} = \frac{-5}{s + 2}$$

4.49. Find $\mathcal{L}\{f(t)\}$ in each case:

$$(a) f(t) = \begin{cases} -1, & 0 \leq t \leq 4 \\ 1, & t > 4 \end{cases}$$

Method 1: Direct Integration

Using the integral definition of the Laplace transform:

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt$$

Split the integral at $t = 4$:

$$\begin{aligned} \mathcal{L}\{f(t)\} &= \int_0^4 e^{-st} (-1) dt + \int_4^{\infty} e^{-st} (1) dt \\ &= -\left[\frac{e^{-st}}{-s}\right]_0^4 + \left[\frac{e^{-st}}{-s}\right]_4^{\infty} \\ &= \frac{1}{s}(e^{-4s} - 1) + \left(0 - \frac{e^{-4s}}{-s}\right) \\ &= \frac{e^{-4s}}{s} - \frac{1}{s} + \frac{e^{-4s}}{s} \\ &= \frac{2e^{-4s} - 1}{s} \end{aligned}$$

Method 2: Unit Step Function *****

Rewrite $f(t)$ using Heaviside unit step functions $u(t)$:

$$f(t) = -1 + 2u(t - 4)$$

Taking the Laplace transform:

$$\begin{aligned} \mathcal{L}\{f(t)\} &= \mathcal{L}\{-1\} + 2\mathcal{L}\{u(t - 4)\} \\ &= -\frac{1}{s} + \frac{2e^{-4s}}{s} = \frac{2e^{-4s} - 1}{s} \end{aligned}$$

$$(b) f(t) = \begin{cases} t+1, & 0 \leq t < 3 \\ 0, & t \geq 3 \end{cases}$$

Method 1: Direct Integration

$$\begin{aligned} \mathcal{L}\{f(t)\} &= \int_0^3 e^{-st} (t+1) dt + \int_3^\infty e^{-st} (0) dt \\ &= \int_0^3 e^{-st} (t+1) dt \end{aligned}$$

Using integration by parts $\int u dv = uv - \int v du$ with $u = t+1$ and $dv = e^{-st} dt$:

- $du = dt$
- $v = -\frac{e^{-s}}{s}$

$$\begin{aligned} \int e^{-st} (t+1) dt &= -\frac{(t+1)e^{-st}}{s} - \int -\frac{e^{-st}}{s} dt \\ &= -\frac{(t+1)e^{-s}}{s} - \frac{e^{-st}}{s^2} \end{aligned}$$

Evaluating from 0 to 3:

$$\begin{aligned} \mathcal{L}\{f(t)\} &= \left[-\frac{(t+1)e^{-st}}{s} - \frac{e^{-st}}{s^2} \right]_0^3 \\ &= \left(-\frac{4e^{-3s}}{s} - \frac{e^{-3s}}{s^2} \right) - \left(-\frac{1}{s} - \frac{1}{s^2} \right) \\ &= \frac{1}{s} + \frac{1}{s^2} - \frac{4e^{-3s}}{s} - \frac{e^{-3s}}{s^2} \end{aligned}$$

Method 2: Unit Step Function

Express $f(t)$ using unit step functions:

$$f(t) = (t+1) - (t+1)u(t-3)$$

To apply the shifting theorem $\mathcal{L}\{g(t-a)u(t-a)\} = e^{-as} \mathcal{L}\{g(t)\}$, rewrite $t+1$ as $(t-3)+4$:

$$f(t) = (t+1) - [(t-3)+4]u(t-3)$$

Taking the Laplace transform term-by-term:

$$\mathcal{L}\{f(t)\} = \mathcal{L}\{t+1\} - \mathcal{L}\{[(t-3)+4]u(t-3)\}$$

$$\begin{aligned}
&= \left(\frac{1}{s^2} + \frac{1}{s} \right) - e^{-3s} \mathcal{L}\{t + 4\} \\
&= \frac{1}{s^2} + \frac{1}{s} - e^{-3s} \left(\frac{1}{s^2} + \frac{4}{s} \right) \\
&= \frac{1}{s} + \frac{1}{s^2} - \frac{4e^{-3s}}{s} - \frac{e^{-3s}}{s^2}
\end{aligned}$$

$$(c) f(t) = \begin{cases} 0, & 0 \leq t < 2 \\ 1, & 2 \leq t < 4 \\ 0, & t \geq 4 \end{cases}$$

Method 1: Direct Integration

$$\begin{aligned}
\mathcal{L}\{f(t)\} &= \int_2^4 e^{-st} (1) dt \\
&= \left[\frac{e^{-st}}{-s} \right]_2^4 = -\frac{e^{-4s}}{s} - \left(-\frac{e^{-2s}}{s} \right) = \frac{e^{-2s} - e^{-4s}}{s}
\end{aligned}$$

Method 2: Unit Step Function

A rectangular pulse active from $t = 2$ to $t = 4$ is represented as:

$$f(t) = u(t - 2) - u(t - 4)$$

Taking the Laplace transform using $\mathcal{L}\{u(t - a)\} = \frac{e^{-as}}{s}$:

$$\begin{aligned}
\mathcal{L}\{f(t)\} &= \mathcal{L}\{u(t - 2)\} - \mathcal{L}\{u(t - 4)\} \\
&= \frac{e^{-2s}}{s} - \frac{e^{-4s}}{s} = \frac{e^{-2s} - e^{-4s}}{s}
\end{aligned}$$

4.60. Find (a) $\mathcal{L}\{2\mathcal{U}(t - 1) + 3\mathcal{U}(t - 2)\}$, (b) $\mathcal{L}\{t\mathcal{U}(t - 3)\}$ and graph each of the given functions of t .

Part (a)

1. Mathematical Evaluation

Let $f(t) = 2\mathcal{U}(t - 1) + 3\mathcal{U}(t - 2)$.

Using the property $\mathcal{L}\{u(t - a)\} = \frac{e^{-as}}{s}$:

$$\begin{aligned}\mathcal{L}\{f(t)\} &= 2\mathcal{L}\{\mathcal{U}(t-1)\} + 3\mathcal{L}\{\mathcal{U}(t-2)\} \\ &= 2\left(\frac{e^{-s}}{s}\right) + 3\left(\frac{e^{-2s}}{s}\right) = \frac{2e^{-s} + 3e^{-2s}}{s}\end{aligned}$$

2. Graphing $f(t)$

The piecewise representation of $f(t)$ is:

$$f(t) = \begin{cases} 0, & 0 \leq t < 1 \\ 2, & 1 \leq t < 2 \\ 5, & t \geq 2 \end{cases}$$

Graph Description:

- $0 \leq t < 1$: A horizontal line along the t -axis at $y = 0$.
- $1 \leq t < 2$: A step jump up to $y = 2$, holding constant until $t = 2$.
- $t \geq 2$: Another step jump up to $y = 5$, extending horizontally to the right for all $t \geq 2$.

Part (b)

1. Mathematical Evaluation

Let $g(t) = t\mathcal{U}(t-3)$.

To apply the Second Translation Theorem $\mathcal{L}\{F(t-a)\mathcal{U}(t-a)\} = e^{-as}f(s)$, rewrite t in terms of $(t-3)$:

$$g(t) = [(t-3) + 3]\mathcal{U}(t-3) = (t-3)\mathcal{U}(t-3) + 3\mathcal{U}(t-3)$$

Taking the Laplace transform term-by-term:

$$\begin{aligned}\mathcal{L}\{g(t)\} &= \mathcal{L}\{(t-3)\mathcal{U}(t-3)\} + 3\mathcal{L}\{\mathcal{U}(t-3)\} \\ &= e^{-3s}\mathcal{L}\{t\} + 3\left(\frac{e^{-3s}}{s}\right) \\ &= e^{-3s}\left(\frac{1}{s^2}\right) + \frac{3e^{-3s}}{s} \\ &= e^{-3s}\left(\frac{1}{s^2} + \frac{3s}{s^2}\right) = \frac{e^{-3s}(3s+1)}{s^2}\end{aligned}$$

2. Graphing $g(t)$

The piecewise representation of $g(t)$ is:

$$g(t) = \begin{cases} 0, & 0 \leq t < 3 \\ t, & t \geq 3 \end{cases}$$

Graph Description:

- $0 \leq t < 3$: A horizontal line along the t -axis at $y = 0$.
- $t \geq 3$: Starting with a step jump at $t = 3$ to $y = 3$, it follows a line with slope $+1$ ($y = t$) extending upwards to the right.

4.63. Given the function $f(t) = \begin{cases} \cos t, & t > \pi/2 \\ 3, & t < \pi/2 \end{cases}$:

(a) Express the function $f(t)$ in terms of the unit step function.

Using the unit step function $\mathcal{U}(t - a)$, a function that transitions from $g(t)$ to $h(t)$ at $t = a$ is given by:

$$f(t) = g(t) + [h(t) - g(t)]\mathcal{U}(t - a)$$

Here, for $t < \pi/2$, $f(t) = 3$, and for $t > \pi/2$, $f(t) = \cos t$:

$$f(t) = 3 + (\cos t - 3)\mathcal{U}\left(t - \frac{\pi}{2}\right)$$

Final Answer (a):

$$3 + (\cos t - 3)\mathcal{U}\left(t - \frac{\pi}{2}\right)$$

(b) Find its Laplace transform $\mathcal{L}\{f(t)\}$.

Expand the expression for $f(t)$:

$$f(t) = 3 + \cos t \cdot \mathcal{U}\left(t - \frac{\pi}{2}\right) - 3\mathcal{U}\left(t - \frac{\pi}{2}\right)$$

To apply the second shifting theorem $\mathcal{L}\{g(t - a)\mathcal{U}(t - a)\} = e^{-as}\mathcal{L}\{g(t)\}$, rewrite $\cos t$ in terms of $\left(t - \frac{\pi}{2}\right)$ using trigonometric identities:

$$\cos t = \cos \left(\left(t - \frac{\pi}{2} \right) + \frac{\pi}{2} \right) = -\sin \left(t - \frac{\pi}{2} \right)$$

Substitute this back into $f(t)$:

$$f(t) = 3 - \sin \left(t - \frac{\pi}{2} \right) u \left(t - \frac{\pi}{2} \right) - 3u \left(t - \frac{\pi}{2} \right)$$

Take the Laplace transform term-by-term:

$$\mathcal{L}\{f(t)\} = \mathcal{L}\{3\} - \mathcal{L}\left\{\sin \left(t - \frac{\pi}{2} \right) u \left(t - \frac{\pi}{2} \right)\right\} - 3\mathcal{L}\left\{u \left(t - \frac{\pi}{2} \right)\right\}$$

Apply standard Laplace transforms:

- $\mathcal{L}\{3\} = \frac{3}{s}$
- $\mathcal{L}\left\{\sin \left(t - \frac{\pi}{2} \right) u \left(t - \frac{\pi}{2} \right)\right\} = e^{-\pi s/2} \mathcal{L}\{\sin t\} = e^{-\pi/2} \left(\frac{1}{s^2 + 1} \right)$
- $3\mathcal{L}\left\{u \left(t - \frac{\pi}{2} \right)\right\} = 3 \cdot \frac{e^{-\pi s/2}}{s}$

Combine the terms:

$$\begin{aligned} \mathcal{L}\{f(t)\} &= \frac{3}{s} - \frac{3e^{-\pi s/2}}{s} - \frac{e^{-\pi/2}}{s^2 + 1} \\ &= \frac{3(1 - e^{-\pi s/2})}{s} - \frac{e^{-\pi s/2}}{s^2 + 1} \end{aligned}$$

Final Answer (b):

$$\frac{3(1 - e^{-\pi s/2})}{s} - \frac{e^{-\pi s/2}}{s^2 + 1}$$

Lecture 15

1. Integration Property

If $\mathcal{L}\{f(t)\} = F(s)$, then:

$$\mathcal{L}\left\{\int_0^t f(u) du\right\} = \frac{F(s)}{s}$$

- **Given Expression:** $\int_0^t (u^2 - u + e^{-u}) du$

- Let $f(t) = t^2 - t + e^{-t}$.
- Taking the Laplace transform:

$$F(s) = \mathcal{L}\{t^2 - t + e^{-t}\} = \frac{2}{s^3} - \frac{1}{s^2} + \frac{1}{s+1}$$

- Applying the integration property:

$$\mathcal{L}\left\{\int_0^t (u^2 - u + e^{-u}) du\right\} = \frac{1}{s} \left(\frac{2}{s^3} - \frac{1}{s^2} + \frac{1}{s+1} \right)$$

2. Multiplication by t^n Property

$$\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} F(s)$$

Problem: Find $\mathcal{L}\{t^2 e^{-3t}\}$

- Let $f(t) = e^{-3t}$, then $F(s) = \mathcal{L}\{e^{-3t}\} = \frac{1}{s+3}$.
- Applying the formula for $n = 2$:

$$\mathcal{L}\{t^2 e^{-3t}\} = (-1)^2 \frac{d^2}{ds^2} \left(\frac{1}{s+3} \right)$$

$$\frac{d}{ds} \left(\frac{1}{s+3} \right) = -\frac{1}{(s+3)^2}$$

$$\frac{d^2}{ds^2} \left(\frac{1}{s+3} \right) = \frac{2}{(s+3)^3}$$

- **Result:** $\frac{2}{(s+3)^3}$

Evaluate: $\mathcal{L}\{t^2 \sin(2t)\}$

Multiplication Property:

$$\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} F(s)$$

Here, $n = 2$ and $f(t) = \sin 2t$.

Step-by-Step Solution:

1. Find $F(s) = \mathcal{L}\{\sin 2t\}$:

$$F(s) = \frac{2}{s^2 + 4}$$

2. Apply the Property for $n = 2$:

$$\mathcal{L}\{t^2 \sin 2t\} = (-1)^2 \frac{d^2}{ds^2} \left(\frac{2}{s^2 + 4} \right) = \frac{d^2}{ds^2} \left(\frac{2}{s^2 + 4} \right)$$

3. First Derivative:

$$\frac{d}{ds} (2(s^2 + 4)^{-1}) = -2(s^2 + 4)^{-2} \cdot 2s = \frac{-4s}{(s^2 + 4)^2}$$

4. Second Derivative:

Using the quotient rule $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$ on $\frac{-4s}{(s^2 + 4)^2}$:

- $u = -4s \Rightarrow u' = -4$
- $v = (s^2 + 4)^2 \Rightarrow v' = 2(s^2 + 4) \cdot 2s = 4s(s^2 + 4)$

$$\frac{d^2}{ds^2} \left(\frac{2}{s^2 + 4} \right) = \frac{-4(s^2 + 4)^2 - (-4s) \cdot 4s(s^2 + 4)}{(s^2 + 4)^4}$$

Factor out $(s^2 + 4)$ from the numerator:

$$\begin{aligned} &= \frac{(s^2 + 4)[-4(s^2 + 4) + 16s^2]}{(s^2 + 4)^4} \\ &= \frac{-4s^2 - 16 + 16s^2}{(s^2 + 4)^3} \\ &= \frac{12s^2 - 16}{(s^2 + 4)^3} \end{aligned}$$

Final Answer:

$$\mathcal{L}\{t^2 \sin 2t\} = \frac{4(3s^2 - 4)}{(s^2 + 4)^3}$$

Homework: Exercises 4.65, 4.66

3. Division by t Property

$$\mathcal{L}\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(u) du$$

Problem: Find $\mathcal{L}\left\{\frac{1-e^{-t}}{t}\right\}$

- Let $f(t) = 1 - e^{-t}$, then $F(s) = \frac{1}{s} - \frac{1}{s+1}$.
- Applying the property:

$$\begin{aligned}\mathcal{L}\left\{\frac{1-e^{-t}}{t}\right\} &= \int_s^\infty \left(\frac{1}{u} - \frac{1}{u+1}\right) du \\&= [\ln u - \ln(u+1)]_s^\infty = \left[\ln\left(\frac{u}{u+1}\right)\right]_s^\infty \\&= \left[\ln\left(\frac{1}{1+\frac{1}{u}}\right)\right]_s^\infty = \ln(1) - \ln\left(\frac{s}{s+1}\right) \\&= \ln\left(\frac{s+1}{s}\right)\end{aligned}$$

Homework: Evaluate $\mathcal{L}\left\{\frac{e^{-4t}-e^{-3t}}{t}\right\}$

Standard Inverse Laplace Transforms

1. Evaluate: $\mathcal{L}^{-1}\left\{\frac{5}{s+2}\right\}$

$$= 5\mathcal{L}^{-1}\left\{\frac{1}{s-(-2)}\right\} = 5e^{-2t}$$

2. Evaluate: $\mathcal{L}^{-1}\left\{\frac{4s}{s^2+4}\right\}$

$$= 4\mathcal{L}^{-1}\left\{\frac{s}{s^2+2^2}\right\} = 4\cos(2t)$$

3. Evaluate:

$$\text{Evaluate: } \mathcal{L}^{-1}\left\{\frac{5s+4}{s^3}\right\}$$

Step-by-Step Solution

1. **Split the Fraction:** Divide each term in the numerator by the denominator s^3 :

$$\mathcal{L}^{-1}\left\{\frac{5s+4}{s^3}\right\} = \mathcal{L}^{-1}\left\{\frac{5s}{s^3} + \frac{4}{s^3}\right\} = \mathcal{L}^{-1}\left\{\frac{5}{s^2} + \frac{4}{s^3}\right\}$$

2. **Apply Linearity of the Inverse Laplace Transform:**

$$\mathcal{L}^{-1}\left\{\frac{5}{s^2}\right\} + \mathcal{L}^{-1}\left\{\frac{4}{s^3}\right\} = 5\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} + 4\mathcal{L}^{-1}\left\{\frac{1}{s^3}\right\}$$

3. **Use the Standard Inverse Transform Formula:** Recall that $\mathcal{L}^{-1}\left\{\frac{1}{s^{n+1}}\right\} = \frac{t^n}{n!}$:

- For $\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\}$ ($n = 1$):

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} = \frac{t^1}{1!} = t$$

- For $\mathcal{L}^{-1}\left\{\frac{1}{s^3}\right\}$ ($n = 2$):

$$\mathcal{L}^{-1}\left\{\frac{1}{s^3}\right\} = \frac{t^2}{2!} = \frac{t^2}{2}$$

4. **Substitute and Simplify:**

$$5(t) + 4\left(\frac{t^2}{2}\right) = 5t + 2t^2$$

Final Answer:

$$\mathcal{L}^{-1}\left\{\frac{5s+4}{s^3}\right\} = 5t + 2t^2$$

4. Evaluate: $\mathcal{L}^{-1}\left\{\frac{4s-3}{s^2+4}\right\}$

$$= 4\mathcal{L}^{-1}\left\{\frac{s}{s^2+4}\right\} - 3\mathcal{L}^{-1}\left\{\frac{1}{s^2+4}\right\}$$

$$= 4\cos(2t) - \frac{3}{2}\sin(2t)$$

First Shifting Theorem in Inverse Transforms

$$\mathcal{L}^{-1}\{F(s-a)\} = e^{at}f(t)$$

Example 1:

Find $\mathcal{L}^{-1}\left\{\frac{3s-4}{(2s-3)^5}\right\}$

- Factoring 2 out of the denominator:

$$\begin{aligned}\mathcal{L}^{-1}\left\{\frac{3s-4}{2^5\left(s-\frac{3}{2}\right)^5}\right\} &= \frac{1}{32}\mathcal{L}^{-1}\left\{\frac{3\left(s-\frac{3}{2}\right)+\frac{9}{2}-4}{\left(s-\frac{3}{2}\right)^5}\right\} \\ &= \frac{1}{32}\mathcal{L}^{-1}\left\{\frac{3\left(s-\frac{3}{2}\right)+\frac{1}{2}}{\left(s-\frac{3}{2}\right)^5}\right\}\end{aligned}$$

- Applying the shifting theorem ($a = 3/2$):

$$\begin{aligned}&= \frac{e^{\frac{3}{2}t}}{32}\mathcal{L}^{-1}\left\{\frac{3}{s^4} + \frac{1/2}{s^5}\right\} \\ &= \frac{e^{\frac{3}{2}t}}{32}\left[3\frac{t^3}{3!} + \frac{1}{2}\frac{t^4}{4!}\right] = \frac{e^{\frac{3}{2}t}}{32}\left(\frac{t^3}{2} + \frac{t^4}{48}\right)\end{aligned}$$

Example 2:

Find $\mathcal{L}^{-1}\left\{\frac{s}{s^2-4s+2}\right\} = \mathcal{L}^{-1}\left\{\frac{s}{(s-2)^2+16}\right\}$

$$\begin{aligned}&= \mathcal{L}^{-1}\left\{\frac{(s-2)+2}{(s-2)^2+16}\right\} \\ &= e^{2t}\mathcal{L}^{-1}\left\{\frac{s+2}{s^2+16}\right\} = e^{2t}\left[\mathcal{L}^{-1}\left\{\frac{s}{s^2+4^2}\right\} + \frac{2}{4}\mathcal{L}^{-1}\left\{\frac{4}{s^2+4^2}\right\}\right] \\ &= e^{2t}\left(\cos(4t) + \frac{1}{2}\sin(4t)\right)\end{aligned}$$

Example 3:

$$\begin{aligned}
\text{Find } \mathcal{L}^{-1}\left\{\frac{3s-4}{s^2-2s-3}\right\} &= \mathcal{L}^{-1}\left\{\frac{3(s-1)-1}{(s-1)^2-4}\right\} \\
&= e^t \mathcal{L}^{-1}\left\{\frac{3s-1}{s^2-2^2}\right\} \\
&= e^t \left[3 \cosh(2t) - \frac{1}{2} \sinh(2t) \right]
\end{aligned}$$

Homework: Exercises 4.12, 4.13, 4.33, 4.53, 4.77, 4.78

Derivatives of Transforms & Partial Fractions

1. Inverse Transform of Logarithmic Function

Property: $\mathcal{L}^{-1}\{F'(s)\} = -tf(t) \Rightarrow f(t) = -\frac{1}{t} \mathcal{L}^{-1}\{F'(s)\}$

Problem: Find $\mathcal{L}^{-1}\left\{\ln\left(1 + \frac{1}{s}\right)\right\}$

- Let $F(s) = \ln\left(1 + \frac{1}{s}\right) = \ln(s+1) - \ln(s)$.
- Differentiating with respect to s :

$$F'(s) = \frac{1}{s+1} - \frac{1}{s}$$

- Taking the inverse transform of $F'(s)$:

$$\mathcal{L}^{-1}\{F'(s)\} = e^{-t} - 1$$

- Applying the derivative relation:

$$-tf(t) = e^{-t} - 1 \Rightarrow f(t) = \frac{1 - e^{-t}}{t}$$

Homework: Exercise 4.79

2. Partial Fractions Method

Problem: Find $\mathcal{L}^{-1} \left\{ \frac{2s^2 - 4}{(s-2)(s+1)(s-3)} \right\}$

Partial Fraction Expansion:

$$\frac{2s^2 - 4}{(s-2)(s+1)(s-3)} = \frac{A}{s-2} + \frac{B}{s+1} + \frac{C}{s-3}$$

$$2s^2 - 4 = A(s+1)(s-3) + B(s-2)(s-3) + C(s-2)(s+1)$$

- **Finding A** (Set $s = 2$):

$$2(2)^2 - 4 = A(2+1)(2-3) \Rightarrow 4 = -3A \Rightarrow A = -\frac{4}{3}$$

- **Finding B** (Set $s = -1$):

$$2(-1)^2 - 4 = B(-1-2)(-1-3) \Rightarrow -2 = 12B \Rightarrow B = -\frac{1}{6}$$

- **Finding C** (Set $s = 3$):

$$2(3)^2 - 4 = C(3-2)(3+1) \Rightarrow 14 = 4C \Rightarrow C = \frac{7}{2}$$

Taking the Inverse Transform:

$$\begin{aligned} \mathcal{L}^{-1} \left\{ \frac{2s^2 - 4}{(s-2)(s+1)(s-3)} \right\} &= \mathcal{L}^{-1} \left\{ \frac{-4/3}{s-2} + \frac{-1/6}{s+1} + \frac{7/2}{s-3} \right\} \\ &= -\frac{4}{3} e^{2t} - \frac{1}{6} e^{-t} + \frac{7}{2} e^{3t} \end{aligned}$$

Homework: Exercises 4.40, 4.41

HOMEWORK

Question

Find the Laplace transform of the given function using the division by t property:

$$\text{Evaluate: } \mathcal{L}\left\{\frac{e^{-4t} - e^{-3t}}{t}\right\}$$

Step-by-Step Solution

1. **Identify $f(t)$ and its Laplace transform $F(s)$:** Let $f(t) = e^{-4t} - e^{-3t}$.

$$F(s) = \mathcal{L}\{f(t)\} = \mathcal{L}\{e^{-4t}\} - \mathcal{L}\{e^{-3t}\} = \frac{1}{s+4} - \frac{1}{s+3}$$

2. **Apply the Division by t Property:**

$$\begin{aligned}\mathcal{L}\left\{\frac{f(t)}{t}\right\} &= \int_s^\infty F(u) du \\ \mathcal{L}\left\{\frac{e^{-4t} - e^{-3t}}{t}\right\} &= \int_s^\infty \left(\frac{1}{u+4} - \frac{1}{u+3}\right) du\end{aligned}$$

3. **Evaluate the Integral:**

$$= [\ln(u+4) - \ln(u+3)]_s^\infty = \left[\ln\left(\frac{u+4}{u+3}\right)\right]_s^\infty$$

4. **Evaluate at the Upper Limit (∞):**

Divide the numerator and denominator by u :

$$\lim_{u \rightarrow \infty} \ln\left(\frac{1 + \frac{4}{u}}{1 + \frac{3}{u}}\right) = \ln\left(\frac{1+0}{1+0}\right) = \ln(1) = 0$$

5. **Subtract the Lower Limit (s):**

$$= 0 - \ln\left(\frac{s+4}{s+3}\right) = \ln\left(\frac{s+3}{s+4}\right)$$

Final Answer:

$$\mathcal{L}\left\{\frac{e^{-4t} - e^{-3t}}{t}\right\} = \ln\left(\frac{s+3}{s+4}\right)$$

Question

Evaluate the following inverse Laplace transform:

$$\mathcal{L}^{-1}\left\{\frac{5s+4}{s^3} - \frac{2s-18}{s^2+9} + \frac{24-30\sqrt{s}}{s^4}\right\}$$

Step-by-Step Solution

1. Break Down into Three Separate Terms (Linearity Property)

$$\mathcal{L}^{-1}\left\{\frac{5s+4}{s^3} - \frac{2s-18}{s^2+9} + \frac{24-30\sqrt{s}}{s^4}\right\} = I_1 - I_2 + I_3$$

2. Evaluate First Term (I_1)

$$I_1 = \mathcal{L}^{-1}\left\{\frac{5s+4}{s^3}\right\} = \mathcal{L}^{-1}\left\{\frac{5}{s^2} + \frac{4}{s^3}\right\}$$

Using $\mathcal{L}^{-1}\left\{\frac{1}{s^{n+1}}\right\} = \frac{t^n}{n!}$:

$$I_1 = 5t + 4 \cdot \frac{t^2}{2!} = \mathbf{5t + 2t^2}$$

3. Evaluate Second Term (I_2)

$$I_2 = \mathcal{L}^{-1}\left\{\frac{2s-18}{s^2+9}\right\} = \mathcal{L}^{-1}\left\{\frac{2s}{s^2+3^2}\right\} - \mathcal{L}^{-1}\left\{\frac{18}{s^2+3^2}\right\}$$

$$I_2 = 2\mathcal{L}^{-1}\left\{\frac{s}{s^2+3^2}\right\} - 6\mathcal{L}^{-1}\left\{\frac{3}{s^2+3^2}\right\}$$

$$I_2 = \mathbf{2 \cos(3t) - 6 \sin(3t)}$$

4. Evaluate Third Term (I_3)

$$I_3 = \mathcal{L}^{-1}\left\{\frac{24-30\sqrt{s}}{s^4}\right\} = \mathcal{L}^{-1}\left\{\frac{24}{s^4}\right\} - \mathcal{L}^{-1}\left\{\frac{30s^{1/2}}{s^4}\right\}$$

$$I_3 = 24\mathcal{L}^{-1}\left\{\frac{1}{s^4}\right\} - 30\mathcal{L}^{-1}\left\{\frac{1}{s^{7/2}}\right\}$$

- For $\mathcal{L}^{-1}\left\{\frac{1}{s^4}\right\}$:

$$\mathcal{L}^{-1}\left\{\frac{1}{s^4}\right\} = \frac{t^3}{3!} = \frac{t^3}{6} \Rightarrow 24 \cdot \frac{t^3}{6} = 4t^3$$

- For $\mathcal{L}^{-1}\left\{\frac{1}{s^{7/2}}\right\}$: Using $\mathcal{L}^{-1}\left\{\frac{1}{s^{p+1}}\right\} = \frac{t^p}{\Gamma(p+1)}$ where $p = \frac{5}{2}$:

$$\Gamma\left(\frac{7}{2}\right) = \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi} = \frac{15\sqrt{\pi}}{8}$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s^{7/2}}\right\} = \frac{t^{5/2}}{\frac{15\sqrt{\pi}}{8}} = \frac{8t^{5/2}}{15\sqrt{\pi}}$$

$$30 \cdot \frac{8t^{5/2}}{15\sqrt{\pi}} = \frac{16t^{5/2}}{\sqrt{\pi}}$$

Combining both parts of I_3 :

$$I_3 = 4t^3 - \frac{16t^{5/2}}{\sqrt{\pi}}$$

5. Combine All Parts

$$I_1 - I_2 + I_3 = (5t + 2t^2) - (2 \cos(3t) - 6 \sin(3t)) + \left(4t^3 - \frac{16t^{5/2}}{\sqrt{\pi}}\right)$$

Final Answer:

$$\begin{aligned} \mathcal{L}^{-1}\left\{\frac{5s+4}{s^3} - \frac{2s-18}{s^2+9} + \frac{24-30\sqrt{s}}{s^4}\right\} \\ = 5t + 2t^2 - 2 \cos(3t) + 6 \sin(3t) + 4t^3 - \frac{16t^{5/2}}{\sqrt{\pi}} \end{aligned}$$

Multiplication by t^n Property:

$$\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} F(s)$$

Exercise 4.65 (a): Find $\mathcal{L}\{t \sin 3t\}$

1. Identify $f(t)$ and $F(s)$:

- $f(t) = \sin 3t$
- $F(s) = \mathcal{L}\{\sin 3t\} = \frac{3}{s^2+9}$

2. Apply the Property ($n = 1$):

$$\begin{aligned} \mathcal{L}\{t \sin 3t\} &= (-1)^1 \frac{d}{ds} \left(\frac{3}{s^2+9} \right) \\ &= -\frac{d}{ds} [3(s^2+9)^{-1}] \\ &= -[-3(s^2+9)^{-2} \cdot (2s)] \end{aligned}$$

$$= \frac{6s}{(s^2 + 9)^2}$$

Exercise 4.65 (b): Find $\mathcal{L}\{t^2 \cos 2t\}$

1. Identify $f(t)$ and $F(s)$:

- $f(t) = \cos 2t$
- $F(s) = \mathcal{L}\{\cos 2t\} = \frac{s}{s^2 + 4}$

2. Apply the Property ($n = 2$):

$$\mathcal{L}\{t^2 \cos 2t\} = (-1)^2 \frac{d^2}{ds^2} \left(\frac{s}{s^2 + 4} \right) = \frac{d^2}{ds^2} \left(\frac{s}{s^2 + 4} \right)$$

3. First Derivative:

Using the Quotient Rule $\left(\frac{u}{v}\right)' = \frac{uv' - uv'}{v^2}$:

$$\frac{d}{ds} \left(\frac{s}{s^2 + 4} \right) = \frac{1 \cdot (s^2 + 4) - s \cdot (2s)}{(s^2 + 4)^2} = \frac{4 - s^2}{(s^2 + 4)^2}$$

4. Second Derivative:

Differentiating $\frac{4-s^2}{(s^2+4)^2}$ with respect to s :

- $u = 4 - s^2 \Rightarrow u' = -2s$
- $v = (s^2 + 4)^2 \Rightarrow v' = 2(s^2 + 4) \cdot 2s = 4s(s^2 + 4)$

$$\frac{d^2}{ds^2} \left(\frac{s}{s^2 + 4} \right) = \frac{-2s(s^2 + 4)^2 - (4 - s^2) \cdot 4s(s^2 + 4)}{(s^2 + 4)^4}$$

Factor out $(s^2 + 4)$ from the numerator:

$$\begin{aligned} &= \frac{(s^2 + 4)[-2s(s^2 + 4) - 4s(4 - s^2)]}{(s^2 + 4)^4} \\ &= \frac{-2s^3 - 8s - 16s + 4s^3}{(s^2 + 4)^3} \\ &= \frac{2s^3 - 24s}{(s^2 + 4)^3} \end{aligned}$$

Exercise 4.66: Find $\mathcal{L}\{te^t \sin t\}$

Method 1: Using Multiplication by t First, then First Shifting Theorem

1. Find $\mathcal{L}\{t \sin t\}$:

- $\mathcal{L}\{\sin t\} = \frac{1}{s^2 + 1}$

$$\circ \quad \mathcal{L}\{t \sin t\} = -\frac{d}{ds} \left(\frac{1}{s^2+1} \right) = \frac{2s}{(s^2+1)^2}$$

2. **Apply First Shifting Theorem** ($\mathcal{L}\{e^{at}g(t)\} = G(s-a)$ where $a = 1$):

$$\begin{aligned} \mathcal{L}\{e^t(t \sin t)\} &= \frac{2s}{(s^2+1)^2} \Big|_{s \rightarrow s-1} \\ &= \frac{2(s-1)}{((s-1)^2+1)^2} \\ &= \frac{2s-2}{(s^2-2s+1+1)^2} \end{aligned}$$

Exercise 4.12

Find:

$$(a) \mathcal{L}^{-1} \left\{ \frac{5}{s+2} \right\}$$

$$(b) \mathcal{L}^{-1} \left\{ \frac{4s-3}{s^2+4} \right\}$$

$$(c) \mathcal{L}^{-1} \left\{ \frac{2s-5}{s^2} \right\}$$

$$(d) \mathcal{L}^{-1} \left\{ \frac{1}{s^k} \right\}, \quad k > 0$$

Solutions:

(a)

$$\mathcal{L}^{-1} \left\{ \frac{5}{s+2} \right\} = 5 \mathcal{L}^{-1} \left\{ \frac{1}{s+2} \right\} = 5e^{-2t}$$

(b)

$$\mathcal{L}^{-1} \left\{ \frac{4s-3}{s^2+4} \right\} = 4 \mathcal{L}^{-1} \left\{ \frac{s}{s^2+4} \right\} - \frac{3}{2} \mathcal{L}^{-1} \left\{ \frac{2}{s^2+4} \right\} = 4 \cos 2t - \frac{3}{2} \sin 2t$$

(c)

$$\mathcal{L}^{-1} \left\{ \frac{2s-5}{s^2} \right\} = 2 \mathcal{L}^{-1} \left\{ \frac{1}{s} \right\} - 5 \mathcal{L}^{-1} \left\{ \frac{1}{s^2} \right\} = 2 - 5t$$

(d)

Since $\mathcal{L}\{t^p\} = \frac{\Gamma(p+1)}{s^{p+1}}$, we have:

$$\mathcal{L} \left\{ \frac{t^p}{\Gamma(p+1)} \right\} = \frac{1}{s^{p+1}} \quad \text{or} \quad \mathcal{L}^{-1} \left\{ \frac{1}{s^{p+1}} \right\} = \frac{t^p}{\Gamma(p+1)}$$

Then letting $p = k - 1$:

$$\mathcal{L}^{-1}\left\{\frac{1}{s^k}\right\} = \frac{t^{k-1}}{\Gamma(k)}$$

Exercise 4.13

Find:

$$(a) \mathcal{L}^{-1}\left\{\frac{4-5s}{s^{3/2}}\right\}$$

$$(b) \mathcal{L}^{-1}\left\{\frac{1}{s^2+2s}\right\}$$

Solutions:

(a)

$$\begin{aligned}\mathcal{L}^{-1}\left\{\frac{4-5s}{s^{3/2}}\right\} &= 4\mathcal{L}^{-1}\left\{\frac{1}{s^{3/2}}\right\} - 5\mathcal{L}^{-1}\left\{\frac{1}{s^{1/2}}\right\} \\ &= 4 \cdot \frac{t^{1/2}}{\Gamma(3/2)} - 5 \cdot \frac{t^{-1/2}}{\Gamma(1/2)}\end{aligned}$$

Since $\Gamma(1/2) = \sqrt{\pi}$ and $\Gamma(3/2) = \frac{1}{2}\sqrt{\pi}$:

$$= \frac{8t^{1/2}}{\sqrt{\pi}} - \frac{5t^{-1/2}}{\sqrt{\pi}} = \frac{8t^{1/2} - 5t^{-1/2}}{\sqrt{\pi}}$$

(b)

$$\begin{aligned}\mathcal{L}^{-1}\left\{\frac{1}{s^2+2s}\right\} &= \mathcal{L}^{-1}\left\{\frac{1}{s(s+2)}\right\} = \frac{1}{2}\mathcal{L}^{-1}\left\{\frac{1}{s} - \frac{1}{s+2}\right\} \\ &= \frac{1}{2}\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} - \frac{1}{2}\mathcal{L}^{-1}\left\{\frac{1}{s+2}\right\} \\ &= \frac{1}{2} - \frac{1}{2}e^{-2t} = \frac{1}{2}(1 - e^{-2t})\end{aligned}$$

Exercise 4.33

Find:

$$(a) \mathcal{L}^{-1}\left\{\frac{2s+3}{s^2-2s+5}\right\}$$

$$(b) \mathcal{L}^{-1}\left\{\frac{1}{\sqrt{s+3}}\right\}$$

$$(c) \mathcal{L}^{-1}\left\{\frac{3s-4}{(2s-3)^5}\right\}$$

Solutions:

(a)

Complete the square in the denominator: $s^2 - 2s + 5 = (s - 1)^2 + 4$.

$$\mathcal{L}^{-1}\left\{\frac{2s+3}{s^2-2s+5}\right\} = \mathcal{L}^{-1}\left\{\frac{2(s-1)+5}{(s-1)^2+4}\right\} = 2\mathcal{L}^{-1}\left\{\frac{s-1}{(s-1)^2+4}\right\} + \frac{5}{2}\mathcal{L}^{-1}\left\{\frac{2}{(s-1)^2+4}\right\}$$

Now, since:

$$\mathcal{L}^{-1}\left\{\frac{s}{s^2+4}\right\} = \cos 2t \quad \text{and} \quad \mathcal{L}^{-1}\left\{\frac{2}{s^2+4}\right\} = \sin 2t$$

By the First Shifting Theorem (Theorem 4-5):

$$\mathcal{L}^{-1}\left\{\frac{s-1}{(s-1)^2+4}\right\} = e^t \cos 2t \quad \text{and} \quad \mathcal{L}^{-1}\left\{\frac{2}{(s-1)^2+4}\right\} = e^t \sin 2t$$

Thus:

$$\mathcal{L}^{-1}\left\{\frac{2s+3}{s^2-2s+5}\right\} = 2e^t \cos 2t + \frac{5}{2}e^t \sin 2t = \frac{1}{2}e^t(4 \cos 2t + 5 \sin 2t)$$

(b)

From Problem 4.12(d), we have $\mathcal{L}^{-1}\left\{\frac{1}{s^k}\right\} = \frac{t^{k-1}}{\Gamma(k)}$.

For $k = \frac{1}{2}$, since $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$:

$$\mathcal{L}^{-1}\left\{\frac{1}{\sqrt{s}}\right\} = \frac{t^{-1/2}}{\sqrt{\pi}}$$

Then by the First Shifting Theorem (Theorem 4-5):

$$\mathcal{L}^{-1}\left\{\frac{1}{\sqrt{s+3}}\right\} = \frac{e^{-3t}t^{-1/2}}{\sqrt{\pi}}$$

(c)

Factor 2 out of the denominator:

$$\mathcal{L}^{-1}\left\{\frac{3s-4}{(2s-3)^5}\right\} = \frac{1}{2^5}\mathcal{L}^{-1}\left\{\frac{3s-4}{(s-3/2)^5}\right\} = \frac{1}{32}\mathcal{L}^{-1}\left\{\frac{3(s-3/2)+1/2}{(s-3/2)^5}\right\}$$

Split into two terms:

$$= \frac{3}{32}\mathcal{L}^{-1}\left\{\frac{1}{(s-3/2)^4}\right\} + \frac{1}{64}\mathcal{L}^{-1}\left\{\frac{1}{(s-3/2)^5}\right\}$$

Apply the shifting theorem $\mathcal{L}^{-1}\left\{\frac{1}{(s-a)^{n+1}}\right\} = \frac{t^n}{n!}e^{at}$:

$$\begin{aligned}
&= \frac{3}{32} \cdot \frac{t^3}{3!} e^{3t/2} + \frac{1}{64} \cdot \frac{t^4}{4!} e^{3t/2} \\
&= \frac{t^3}{64} e^{3t/2} + \frac{t^4}{1536} e^{3t/2} \\
&= \frac{t^3(t+8)e^{3t/2}}{1536}
\end{aligned}$$

Exercise 4.53 (a) through (j)

(a) Find $\mathcal{L}^{-1}\left\{\frac{2}{s-3}\right\}$

$$\mathcal{L}^{-1}\left\{\frac{2}{s-3}\right\} = 2\mathcal{L}^{-1}\left\{\frac{1}{s-3}\right\} = 2e^{3t}$$

(b) Find $\mathcal{L}^{-1}\left\{\frac{1}{3s+5}\right\}$

$$\mathcal{L}^{-1}\left\{\frac{1}{3\left(s+\frac{5}{3}\right)}\right\} = \frac{1}{3}\mathcal{L}^{-1}\left\{\frac{1}{s-\left(-\frac{5}{3}\right)}\right\} = \frac{1}{3}e^{-5t/3}$$

(c) Find $\mathcal{L}^{-1}\left\{\frac{4s}{s^2+16}\right\}$

$$\mathcal{L}^{-1}\left\{\frac{4s}{s^2+4^2}\right\} = 4\mathcal{L}^{-1}\left\{\frac{s}{s^2+4^2}\right\} = 4\cos 4t$$

(d) Find $\mathcal{L}^{-1}\left\{\frac{1}{4s^2+9}\right\}$

$$\mathcal{L}^{-1}\left\{\frac{1}{4\left(s^2+\frac{9}{4}\right)}\right\} = \frac{1}{4}\mathcal{L}^{-1}\left\{\frac{1}{s^2+\left(\frac{3}{2}\right)^2}\right\}$$

Multiply and divide by $\frac{3}{2}$:

$$= \frac{1}{4} \cdot \frac{2}{3} \mathcal{L}^{-1}\left\{\frac{3/2}{s^2+\left(\frac{3}{2}\right)^2}\right\} = \frac{1}{6}\sin\left(\frac{3}{2}t\right)$$

(e) Find $\mathcal{L}^{-1}\left\{\frac{2s-8}{s^2+36}\right\}$

$$\begin{aligned}
\mathcal{L}^{-1}\left\{\frac{2s}{s^2+6^2}\right\} - \mathcal{L}^{-1}\left\{\frac{8}{s^2+6^2}\right\} &= 2\mathcal{L}^{-1}\left\{\frac{s}{s^2+6^2}\right\} - \frac{8}{6}\mathcal{L}^{-1}\left\{\frac{6}{s^2+6^2}\right\} \\
&= 2\cos 6t - \frac{4}{3}\sin 6t
\end{aligned}$$

(f) Find $\mathcal{L}^{-1}\left\{\frac{s^3-s^2+s-1}{s^5}\right\}$

Split the fraction term by term:

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2} - \frac{1}{s^3} + \frac{1}{s^4} - \frac{1}{s^5}\right\}$$

Using $\mathcal{L}^{-1}\left\{\frac{1}{s^{n+1}}\right\} = \frac{t^n}{n!}$:

$$= \frac{t^1}{1!} - \frac{t^2}{2!} + \frac{t^3}{3!} - \frac{t^4}{4!} = \mathbf{t - \frac{t^2}{2} + \frac{t^3}{6} - \frac{t^4}{24}}$$

(g) Find $\mathcal{L}^{-1}\left\{\frac{2-s}{s^{3/2}}\right\}$

$$\mathcal{L}^{-1}\left\{\frac{2}{s^{3/2}} - \frac{1}{s^{1/2}}\right\} = 2\mathcal{L}^{-1}\left\{\frac{1}{s^{3/2}}\right\} - \mathcal{L}^{-1}\left\{\frac{1}{s^{1/2}}\right\}$$

Using $\mathcal{L}^{-1}\left\{\frac{1}{s^{p+1}}\right\} = \frac{t^p}{\Gamma(p+1)}$:

- For $s^{3/2}$ ($p = 1/2$): $\Gamma(3/2) = \frac{\sqrt{\pi}}{2} \Rightarrow \mathcal{L}^{-1}\left\{\frac{1}{s^{3/2}}\right\} = \frac{2t^{1/2}}{\sqrt{\pi}}$

- For $s^{1/2}$ ($p = -1/2$): $\Gamma(1/2) = \sqrt{\pi} \Rightarrow \mathcal{L}^{-1}\left\{\frac{1}{s^{1/2}}\right\} = \frac{t^{-1/2}}{\sqrt{\pi}}$

$$= 2\left(\frac{2t^{1/2}}{\sqrt{\pi}}\right) - \frac{t^{-1/2}}{\sqrt{\pi}} = \frac{\mathbf{4t^{1/2} - t^{-1/2}}}{\sqrt{\pi}}$$

(h) Find $\mathcal{L}^{-1}\left\{\frac{3s-16}{s^2-64}\right\}$

$$\mathcal{L}^{-1}\left\{\frac{3s}{s^2-8^2}\right\} - \mathcal{L}^{-1}\left\{\frac{16}{s^2-8^2}\right\} = 3\mathcal{L}^{-1}\left\{\frac{s}{s^2-8^2}\right\} - 2\mathcal{L}^{-1}\left\{\frac{8}{s^2-8^2}\right\}$$

$$= \mathbf{3 \cosh 8 t - 2 \sinh 8 t}$$

(i) Find $\mathcal{L}^{-1}\left\{\frac{(2s+1)^2}{s^5}\right\}$

Expand the numerator:

$$\mathcal{L}^{-1}\left\{\frac{4s^2 + 4s + 1}{s^5}\right\} = \mathcal{L}^{-1}\left\{\frac{4}{s^3} + \frac{4}{s^4} + \frac{1}{s^5}\right\}$$

$$= 4\left(\frac{t^2}{2!}\right) + 4\left(\frac{t^3}{3!}\right) + \frac{t^4}{4!} = \mathbf{2t^2 + \frac{2}{3}t^3 + \frac{t^4}{24}}$$

(j) Find $\mathcal{L}^{-1}\left\{\frac{s}{(s+3)(s+5)}\right\}$

Using Partial Fraction Decomposition:

$$\frac{s}{(s+3)(s+5)} = \frac{A}{s+3} + \frac{B}{s+5}$$

$$s = A(s+5) + B(s+3)$$

- For $s = -3$: $-3 = A(2) \Rightarrow A = -\frac{3}{2}$
- For $s = -5$: $-5 = B(-2) \Rightarrow B = \frac{5}{2}$

$$\mathcal{L}^{-1}\left\{-\frac{3/2}{s+3} + \frac{5/2}{s+5}\right\} = \frac{5}{2}e^{-5t} - \frac{3}{2}e^{-3t}$$

Exercise 4.77

(a) Find $\mathcal{L}^{-1}\left\{\frac{1}{s^2-4}\right\}$

Using standard transform pair $\mathcal{L}^{-1}\left\{\frac{a}{s^2-a^2}\right\} = \sinh(at)$ for $a = 2$:

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2-4}\right\} = \frac{1}{2}\mathcal{L}^{-1}\left\{\frac{2}{s^2-4}\right\} = \frac{1}{2}\sinh 2t$$

(b) Find $\mathcal{L}^{-1}\left\{\frac{1}{s(s+1)(s+2)}\right\}$

1. Partial Fractions:

$$\frac{1}{s(s+1)(s+2)} = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{s+2}$$

- $A = \lim_{s \rightarrow 0} \frac{1}{(s+1)(s+2)} = \frac{1}{2}$
- $B = \lim_{s \rightarrow -1} \frac{1}{s(s+2)} = -1$
- $C = \lim_{s \rightarrow -2} \frac{1}{s(s+1)} = \frac{1}{2}$

2. Inverse Transform:

$$\mathcal{L}^{-1}\left\{\frac{1/2}{s} - \frac{1}{s+1} + \frac{1/2}{s+2}\right\} = \frac{1}{2} - e^{-t} + \frac{1}{2}e^{-2t}$$

(c) Find $\mathcal{L}^{-1}\left\{\frac{8-10s}{(s+1)(s-2)^2}\right\}$

1. Partial Fractions:

$$\frac{8-10s}{(s+1)(s-2)^2} = \frac{A}{s+1} + \frac{B}{s-2} + \frac{C}{(s-2)^2}$$

- $A = \lim_{s \rightarrow -1} \frac{8-10s}{(s-2)^2} = \frac{18}{9} = 2$
- $C = \lim_{s \rightarrow 2} \frac{8-10s}{s+1} = \frac{-12}{3} = -4$
- Equating numerator constant term: $8 = A(4) - B(2) + C(1) \Rightarrow 8 = 8 - 2B - 4 \Rightarrow B = -2$

2. Inverse Transform:

$$\mathcal{L}^{-1}\left\{\frac{2}{s+1} - \frac{2}{s-2} - \frac{4}{(s-2)^2}\right\} = 2e^{-t} - 2e^{2t} - 4te^{2t}$$

(d) Find $\mathcal{L}^{-1}\left\{\frac{3s-12}{(s^2+8)(s-1)}\right\}$

1. Partial Fractions:

$$\frac{3s-12}{(s^2+8)(s-1)} = \frac{As+B}{s^2+8} + \frac{C}{s-1}$$

- $C = \lim_{s \rightarrow 1} \frac{3s-12}{s^2+8} = \frac{-9}{9} = -1$
- $3s-12 = (As+B)(s-1) - 1(s^2+8) = (A-1)s^2 + (B-A)s + (-B-8)$
- $A-1 = 0 \Rightarrow A = 1$
- $-B-8 = -12 \Rightarrow B = 4$

2. Inverse Transform:

$$\begin{aligned}\mathcal{L}^{-1}\left\{\frac{s+4}{s^2+8} - \frac{1}{s-1}\right\} &= \mathcal{L}^{-1}\left\{\frac{s}{s^2+(2\sqrt{2})^2}\right\} + \sqrt{2}\mathcal{L}^{-1}\left\{\frac{2\sqrt{2}}{s^2+(2\sqrt{2})^2}\right\} - \mathcal{L}^{-1}\left\{\frac{1}{s-1}\right\} \\ &= \cos 2\sqrt{2}t + \sqrt{2}\sin 2\sqrt{2}t - e^t\end{aligned}$$

(e) Find $\mathcal{L}^{-1}\left\{\frac{150}{(s^2+2s+5)(s^2-4s+8)}\right\}$

1. Partial Fractions:

$$\frac{150}{(s^2+2s+5)(s^2-4s+8)} = \frac{As+B}{s^2+2s+5} + \frac{Cs+D}{s^2-4s+8}$$

Solving for coefficients yields: $A = 4, B = 10, C = -4, D = 6$.

2. Complete Squares & Rearrange:

- Term 1: $\frac{4s+10}{(s+1)^2+4} = \frac{4(s+1)+6}{(s+1)^2+4} = 4\frac{s+1}{(s+1)^2+4} + 3\frac{2}{(s+1)^2+4}$
- Term 2: $\frac{-4s+6}{(s-2)^2+4} = \frac{-4(s-2)+2}{(s-2)^2+4} = -4\frac{s-2}{(s-2)^2+4} - 1\frac{2}{(s-2)^2+4}$ (or grouped as $3\sin 2t - 4\cos 2t$)

3. Inverse Transform:

$$= e^{-t}(4\cos 2t + 3\sin 2t) + e^{2t}(3\sin 2t - 4\cos 2t)$$

(f) Find $\mathcal{L}^{-1}\left\{\frac{1}{s^4+4}\right\}$

1. **Factor Denominator:**

$$s^4 + 4 = (s^2 + 2)^2 - 4s^2 = (s^2 + 2s + 2)(s^2 - 2s + 2)$$

2. **Partial Fractions:**

$$\begin{aligned}\frac{1}{s^4 + 4} &= \frac{1}{8} \left[\frac{s + 2}{s^2 + 2s + 2} - \frac{s - 2}{s^2 - 2s + 2} \right] \\ &= \frac{1}{8} \left[\frac{(s + 1) + 1}{(s + 1)^2 + 1} - \frac{(s - 1) - 1}{(s - 1)^2 + 1} \right]\end{aligned}$$

3. **Inverse Transform:**

$$\begin{aligned}&= \frac{1}{8} [e^{-t}(\cos t + \sin t) - e^t(\cos t - \sin t)] \\ &= \frac{1}{4} \left[\left(\frac{e^t + e^{-t}}{2} \right) \sin t - \left(\frac{e^t - e^{-t}}{2} \right) \cos t \right] = \frac{1}{4} (\sin t \cosh t - \cos t \sinh t)\end{aligned}$$

Exercise 4.78

(a) Find $\mathcal{L}^{-1}\left\{\frac{1}{s^4-2s^3}\right\}$

1. **Partial Fractions:**

$$\frac{1}{s^3(s-2)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s^3} + \frac{D}{s-2} = -\frac{1}{8s} - \frac{1}{4s^2} - \frac{1}{2s^3} + \frac{1}{8(s-2)}$$

2. **Inverse Transform:**

$$= -\frac{1}{8} - \frac{1}{4}t - \frac{1}{4}t^2 + \frac{1}{8}e^{2t} = \frac{1}{8}(e^{2t} - 2t^2 - 2t - 1)$$

(b) Find $\mathcal{L}^{-1}\left\{\frac{s^2}{(s^2+1)^2}\right\}$

Using the derivative property $\mathcal{L}\{t \cos t\} = -\frac{d}{ds}\left(\frac{s}{s^2+1}\right) = \frac{s^2-1}{(s^2+1)^2}$ and standard pair

$$\mathcal{L}\{\sin t\} = \frac{1}{s^2+1}:$$

$$\begin{aligned}\frac{s^2}{(s^2+1)^2} &= \frac{1}{2} \left[\frac{1}{s^2+1} + \frac{s^2-1}{(s^2+1)^2} \right] \\ \mathcal{L}^{-1}\left\{\frac{s^2}{(s^2+1)^2}\right\} &= \frac{1}{2} (\sin t + t \cos t)\end{aligned}$$

(c) Find $\mathcal{L}^{-1}\left\{\frac{4s-2}{(s^2+1)^2}\right\}$

Split into two terms:

$$\mathcal{L}^{-1}\left\{\frac{4s}{(s^2+1)^2}\right\} - \mathcal{L}^{-1}\left\{\frac{2}{(s^2+1)^2}\right\}$$

- Since $\mathcal{L}^{-1}\left\{\frac{2s}{(s^2+1)^2}\right\} = t \sin t \Rightarrow \mathcal{L}^{-1}\left\{\frac{4s}{(s^2+1)^2}\right\} = 2t \sin t$
- Since $\mathcal{L}^{-1}\left\{\frac{1}{(s^2+1)^2}\right\} = \frac{1}{2}(\sin t - t \cos t) \Rightarrow \mathcal{L}^{-1}\left\{\frac{2}{(s^2+1)^2}\right\} = \sin t - t \cos t$

Combining both:

$$= 2t \sin t - \sin t + t \cos t$$

(d) Find $\mathcal{L}^{-1}\left\{\frac{1}{(s^2+1)^3}\right\}$

Using repeated differentiation property $\mathcal{L}\{t^2 \sin t\} = \frac{6s^2-2}{(s^2+1)^3}$ alongside lower order terms:

$$\mathcal{L}^{-1}\left\{\frac{1}{(s^2+1)^3}\right\} = \frac{1}{8}(3 \sin t - 3t \cos t - t^2 \sin t)$$

Key Property: Differentiation of Laplace Transform

$$\mathcal{L}\{t \cdot f(t)\} = -\frac{d}{ds}F(s) \Rightarrow \mathcal{L}^{-1}\left\{\frac{d}{ds}F(s)\right\} = -t \cdot f(t)$$

Exercise 4.79: Find $\mathcal{L}^{-1}\left\{\ln\left(\frac{s+6}{s+2}\right)\right\}$

1. **Define $F(s)$ and rewrite using logarithm rules:** Let $F(s) = \ln\left(\frac{s+6}{s+2}\right)$.

$$F(s) = \ln(s+6) - \ln(s+2)$$

2. **Differentiate $F(s)$ with respect to s :**

$$F'(s) = \frac{d}{ds}[\ln(s+6) - \ln(s+2)]$$

$$F'(s) = \frac{1}{s+6} - \frac{1}{s+2}$$

3. **Take the inverse Laplace transform of $F'(s)$:**

$$\mathcal{L}^{-1}\{F'(s)\} = \mathcal{L}^{-1}\left\{\frac{1}{s+6}\right\} - \mathcal{L}^{-1}\left\{\frac{1}{s+2}\right\}$$

$$\mathcal{L}^{-1}\{F'(s)\} = e^{-6t} - e^{-2t}$$

4. **Apply the differentiation property:**

Since $\mathcal{L}^{-1}\{F'(s)\} = -t \cdot f(t)$, we set:

$$-t \cdot f(t) = e^{-6t} - e^{-2t}$$

5. **Solve for $f(t)$:**

$$f(t) = \frac{e^{-6t} - e^{-2t}}{-t}$$

$$f(t) = \frac{e^{-2t} - e^{-6t}}{t}$$

Problem 4.40

Find:

$$\mathcal{L}^{-1}\left\{\frac{3s+1}{(s-1)(s^2+1)}\right\}$$

Solution:

1. **Partial Fraction Decomposition Setup:**

$$\frac{3s+1}{(s-1)(s^2+1)} = \frac{A}{s-1} + \frac{Bs+C}{s^2+1} \quad \text{--- (1)}$$

2. **Clear Fractions:**

$$3s+1 = A(s^2+1) + (Bs+C)(s-1)$$

3. **Determine Coefficients:**

○ Letting $s = 1$:

$$3(1) + 1 = A(1^2 + 1) \Rightarrow 4 = 2A \Rightarrow \mathbf{A = 2}$$

• Letting $s = 0$:

$$1 = A(1) + C(-1) \Rightarrow 1 = 2 - C \Rightarrow \mathbf{C = 1}$$

• Letting $s = -1$:

$$-2 = 2A - 2(C - B) \Rightarrow -2 = 4 - 2(1 - B) \Rightarrow \mathbf{B = -2}$$

> **Another method:** Multiplying (1) by s after finding A and letting $s \rightarrow \infty$, we find $A + B = 0$ or $B = -A = -2$. This method affords some simplification of procedure.

4. **Take the Inverse Laplace Transform:**

$$\begin{aligned}
 \mathcal{L}^{-1}\left\{\frac{3s+1}{(s-1)(s^2+1)}\right\} &= \mathcal{L}^{-1}\left\{\frac{2}{s-1}\right\} + \mathcal{L}^{-1}\left\{\frac{-2s+1}{s^2+1}\right\} \\
 &= 2\mathcal{L}^{-1}\left\{\frac{1}{s-1}\right\} - 2\mathcal{L}^{-1}\left\{\frac{s}{s^2+1}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{s^2+1}\right\} \\
 &= 2e^t - 2\cos t + \sin t
 \end{aligned}$$

Problem 4.41 ****

Find:

$$\mathcal{L}^{-1}\left\{\frac{5s^2 - 15s + 7}{(s+1)(s-2)^3}\right\}$$

Solution:

1. **Partial Fraction Decomposition Setup:**

$$\frac{5s^2 - 15s + 7}{(s+1)(s-2)^3} = \frac{A}{s+1} + \frac{B}{(s-2)^3} + \frac{C}{(s-2)^2} + \frac{D}{s-2} \quad \text{--- (1)}$$

2. **Clear Fractions:**

$$5s^2 - 15s + 7 = A(s-2)^3 + B(s+1) + C(s+1)(s-2) + D(s+1)(s-2)^2$$

3. **Determine Coefficients:**

○ Letting $s = -1$:

$$5(-1)^2 - 15(-1) + 7 = A(-1-2)^3 \Rightarrow 27 = -27A \Rightarrow \mathbf{A = -1}$$

• Letting $s = 2$:

$$5(2)^2 - 15(2) + 7 = B(2+1) \Rightarrow -3 = 3B \Rightarrow \mathbf{B = -1}$$

• Letting $s = 0$:

$$7 = -1(-8) - 1(1) + C(1)(-2) + D(1)(4) \Rightarrow -2C + 4D = 0$$

• Letting $s = 1$:

$$-3 = -1(-1) - 1(2) + C(2)(-1) + D(2)(1) \Rightarrow -2C + 2D = -2$$

• Solving the system $-2C + 4D = 0$ and $-2C + 2D = -2$ yields:

$$\mathbf{C = 2, \quad D = 1}$$

> **Another method:** On multiplying (1) by s , after finding A , and letting $s \rightarrow \infty$, we find that $A + D = 0$ or $D = 1$, providing some simplification in the procedure.

4. **Take the Inverse Laplace Transform:**

$$\begin{aligned}
\mathcal{L}^{-1}\left\{\frac{5s^2-15s+7}{(s+1)(s-2)^3}\right\} &= \mathcal{L}^{-1}\left\{\frac{-1}{s+1} + \frac{-1}{(s-2)^3} + \frac{2}{(s-2)^2} + \frac{1}{s-2}\right\} \\
&= -\mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\} - \mathcal{L}^{-1}\left\{\frac{1}{(s-2)^3}\right\} + 2\mathcal{L}^{-1}\left\{\frac{1}{(s-2)^2}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{s-2}\right\} \\
&= -\mathbf{e}^{-t} - \frac{\mathbf{1}}{2}\mathbf{t}^2\mathbf{e}^{2t} + 2\mathbf{t}\mathbf{e}^{2t} + \mathbf{e}^{2t}
\end{aligned}$$

Lecture - 16

Solved Example

Problem:

Solve the differential equation: $Y''(t) - 3Y'(t) + 2Y(t) = 4e^{2t}$ with initial conditions: $Y(0) = -3$, $Y'(0) = 5$

Solution:

Taking the Laplace transform on both sides of the given differential equation:

$$\mathcal{L}\{Y''(t)\} - 3\mathcal{L}\{Y'(t)\} + 2\mathcal{L}\{Y(t)\} = 4\mathcal{L}\{e^{2t}\}$$

Using the initial conditions:

$$s^2y(s) - sY(0) - Y'(0) - 3(sy(s) - Y(0)) + 2y(s) = 4 \cdot \frac{1}{s-2}$$

$$s^2y(s) + 3s - 5 - 3sy(s) - 9 + 2y(s) = \frac{4}{s-2}$$

$$(s^2 - 3s + 2)y(s) + 3s - 14 = \frac{4}{s-2}$$

$$(s-1)(s-2)y(s) = \frac{4}{s-2} - 3s + 14$$

$$y(s) = \frac{4}{(s-1)(s-2)^2} - \frac{3s}{(s-1)(s-2)} + \frac{14}{(s-1)(s-2)}$$

$$y(s) = \frac{-3s^2 + 20s - 24}{(s-1)(s-2)^2}$$

Using partial fraction decomposition and taking the inverse Laplace transform:

$$\mathcal{L}^{-1}\{y(s)\} = \mathcal{L}^{-1}\left\{\frac{-3s^2 + 20s - 24}{(s-1)(s-2)^2}\right\}$$

$$Y(t) = \mathcal{L}^{-1}\left\{\frac{-7}{s-1} + \frac{4}{s-2} + \frac{4}{(s-2)^2}\right\}$$

$$Y(t) = -7e^t + 4e^{2t} + 4te^{2t}$$

HW Problem 1

Solve:

$$Y'' + 2Y' + 5Y = e^{-t} \sin t, \quad Y(0) = 0, \quad Y'(0) = 1$$

Solution: Taking the Laplace transform on both sides:

$$\mathcal{L}\{Y''\} + 2\mathcal{L}\{Y'\} + 5\mathcal{L}\{Y\} = \mathcal{L}\{e^{-t} \sin t\}$$

$$s^2 y(s) - sY(0) - Y'(0) + 2(sy(s) - Y(0)) + 5y(s) = \frac{1}{(s+1)^2 + 1}$$

Substituting $Y(0) = 0$ and $Y'(0) = 1$:

$$s^2 y(s) - 0 - 1 + 2sy(s) + 5y(s) = \frac{1}{s^2 + 2s + 2}$$

$$(s^2 + 2s + 5)y(s) - 1 = \frac{1}{s^2 + 2s + 2}$$

$$(s^2 + 2s + 5)y(s) = 1 + \frac{1}{s^2 + 2s + 2} = \frac{s^2 + 2s + 3}{s^2 + 2s + 2}$$

$$y(s) = \frac{s^2 + 2s + 3}{(s^2 + 2s + 5)(s^2 + 2s + 2)}$$

Using partial fraction decomposition by letting $u = s^2 + 2s$:

$$\frac{u + 3}{(u + 5)(u + 2)} = \frac{2/3}{u + 5} + \frac{1/3}{u + 2}$$

$$y(s) = \frac{2/3}{s^2 + 2s + 5} + \frac{1/3}{s^2 + 2s + 2}$$

Completing the square in the denominators:

$$y(s) = \frac{1}{3} \cdot \frac{2}{(s+1)^2 + 2^2} + \frac{1}{3} \cdot \frac{1}{(s+1)^2 + 1^2}$$

Taking the inverse Laplace transform:

$$Y(t) = \frac{1}{3} e^{-t} \sin(2t) + \frac{1}{3} e^{-t} \sin(t)$$

$$Y(t) = \frac{1}{3} e^{-t} (\sin t + \sin 2t)$$

HW Problem 2

Solve:

$$Y''' - 3Y'' + 3Y' - Y = t^2 e^t, \quad Y(0) = 1, \quad Y'(0) = 0, \quad Y''(0) = -2$$

Solution: Taking the Laplace transform on both sides:

$$\begin{aligned} \mathcal{L}\{Y'''\} - 3\mathcal{L}\{Y''\} + 3\mathcal{L}\{Y'\} - \mathcal{L}\{Y\} &= \mathcal{L}\{t^2 e^t\} \\ (s^3 y(s) - s^2 Y(0) - sY'(0) - Y''(0)) - 3(s^2 y(s) - sY(0) - Y'(0)) + 3(sy(s) - Y(0)) \\ &\quad - y(s) = \frac{2}{(s-1)^3} \end{aligned}$$

Substituting $Y(0) = 1, Y'(0) = 0, Y''(0) = -2$:

$$\begin{aligned} (s^3 - 3s^2 + 3s - 1)y(s) - s^2 + 2 + 3s - 3 &= \frac{2}{(s-1)^3} \\ (s-1)^3 y(s) - (s^2 - 3s + 1) &= \frac{2}{(s-1)^3} \\ (s-1)^3 y(s) &= \frac{2}{(s-1)^3} + s^2 - 3s + 1 \end{aligned}$$

Dividing by $(s-1)^3$:

$$y(s) = \frac{2}{(s-1)^6} + \frac{s^2 - 3s + 1}{(s-1)^3}$$

Rewriting $s^2 - 3s + 1$ as $(s-1)^2 - (s-1) - 1$:

$$y(s) = \frac{2}{(s-1)^6} + \frac{1}{s-1} - \frac{1}{(s-1)^2} - \frac{1}{(s-1)^3}$$

Taking inverse Laplace transform on both sides:

$$\begin{aligned} Y(t) &= 2 \cdot \frac{t^5}{5!} e^t + e^t - te^t - \frac{t^2}{2} e^t \\ Y(t) &= e^t \left(1 - t - \frac{t^2}{2} + \frac{t^5}{60} \right) \end{aligned}$$

HW Problem

Solve the previously done circuit problems by using Laplace transformation.

HW: 4.42–4.44, 4.83, 4.84

Problem 4.42

Solve the differential equation:

$$y''(t) + y(t) = 1 \quad \text{given } y(0) = 1, y'(0) = 0$$

Solution:

Step 1: Take the Laplace transform of both sides Applying the Laplace transform $\mathcal{L}$ to both sides:

$$\mathcal{L}\{y''(t) + y(t)\} = \mathcal{L}\{1\}$$

Let $Y(s) = \mathcal{L}\{y(t)\}$. Using the transform properties for derivatives and constants:

$$s^2Y(s) - sy(0) - y'(0) + Y(s) = \frac{1}{s}$$

Step 2: Substitute the initial conditions Given $y(0) = 1$ and $y'(0) = 0$:

$$s^2Y(s) - s(1) - 0 + Y(s) = \frac{1}{s}$$

$$s^2Y(s) - s + Y(s) = \frac{1}{s}$$

Step 3: Solve for $Y(s)$ Group the $Y(s)$ terms on the left-hand side:

$$(s^2 + 1)Y(s) = s + \frac{1}{s}$$

$$(s^2 + 1)Y(s) = \frac{s^2 + 1}{s}$$

Divide both sides by $(s^2 + 1)$:

$$Y(s) = \frac{s^2 + 1}{s(s^2 + 1)} = \frac{1}{s}$$

Step 4: Take the Inverse Laplace Transform Taking the inverse Laplace transform on both sides:

$$y(t) = \mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\left\{\frac{1}{s}\right\} = 1$$

Final Answer:

$$y(t) = 1$$

Problem 4.43

Solve:

$$y'' - 3y' + 2y = 2e^{-t}, \quad y(0) = 2, \quad y'(0) = -1$$

Solution:

Step 1: Take the Laplace transform of both sides Taking the Laplace transform of the given differential equation:

$$[s^2Y - sy(0) - y'(0)] - 3[sY - y(0)] + 2Y = \frac{2}{s+1}$$

Step 2: Substitute initial conditions and solve for Y Using $y(0) = 2$ and $y'(0) = -1$:

$$[s^2Y - 2s - (-1)] - 3[sY - 2] + 2Y = \frac{2}{s+1}$$

$$s^2Y - 2s + 1 - 3sY + 6 + 2Y = \frac{2}{s+1}$$

$$(s^2 - 3s + 2)Y - 2s + 7 = \frac{2}{s+1}$$

$$(s^2 - 3s + 2)Y = \frac{2}{s+1} + 2s - 7$$

$$(s^2 - 3s + 2)Y = \frac{2 + (2s - 7)(s + 1)}{s + 1}$$

$$(s^2 - 3s + 2)Y = \frac{2 + (2s^2 - 5s - 7)}{s + 1} = \frac{2s^2 - 5s - 5}{s + 1}$$

Factoring $s^2 - 3s + 2 = (s - 1)(s - 2)$:

$$Y = \frac{2s^2 - 5s - 5}{(s + 1)(s - 1)(s - 2)}$$

Step 3: Perform partial fraction decomposition

$$Y = \frac{A}{s+1} + \frac{B}{s-1} + \frac{C}{s-2}$$

Using the cover-up method:

- For A (set $s = -1$):

$$A = \frac{2(-1)^2 - 5(-1) - 5}{(-1-1)(-1-2)} = \frac{2+5-5}{(-2)(-3)} = \frac{2}{6} = \frac{1}{3}$$

- For B (set $s = 1$):

$$B = \frac{2(1)^2 - 5(1) - 5}{(1+1)(1-2)} = \frac{2-5-5}{(2)(-1)} = \frac{-8}{-2} = 4$$

- For C (set $s = 2$):

$$C = \frac{2(2)^2 - 5(2) - 5}{(2+1)(2-1)} = \frac{8-10-5}{(3)(1)} = \frac{-7}{3}$$

Thus,

$$Y = \frac{1/3}{s+1} + \frac{4}{s-1} + \frac{-7/3}{s-2}$$

Step 4: Take the Inverse Laplace Transform Taking the inverse Laplace transform, we obtain the required solution:

$$y(t) = \mathcal{L}^{-1}\{Y\}$$
$$y = \frac{1}{3}e^{-t} + 4e^t - \frac{7}{3}e^{2t}$$

Problem 4.44

Solve:

$$y^{(iv)} + 2y'' + y = \sin t, \quad y(0) = 1, \quad y'(0) = -2, \quad y''(0) = 3, \quad y'''(0) = 0$$

Solution:

Taking the Laplace transform of the given differential equation and using the initial conditions:

$$[s^4 Y - s^3(1) - s^2(-2) - s(3) - 0] + 2[s^2 Y - s(1) - (-2)] + Y = \frac{1}{s^2 + 1}$$

which can be written as:

$$(s^4 + 2s^2 + 1)Y = \frac{1}{s^2 + 1} + s^3 - 2s^2 + 5s - 4$$

or

$$\begin{aligned} Y &= \frac{1}{(s^2 + 1)^3} + \frac{s^3 - 2s^2 + 5s - 4}{(s^2 + 1)^2} = \frac{1}{(s^2 + 1)^3} + \frac{(s^3 + s) - 2(s^2 + 1) + 4s - 2}{(s^2 + 1)^2} \\ &= \frac{1}{(s^2 + 1)^3} + \frac{s}{s^2 + 1} - \frac{2}{s^2 + 1} + \frac{4s - 2}{(s^2 + 1)^2} \end{aligned}$$

Now by using the special theorems on Laplace transforms [see Problem 4.78]:

$$\mathcal{L}^{-1}\left\{\frac{1}{(s^2 + 1)^3}\right\} = \frac{3}{8}\sin t - \frac{3}{8}t \cos t - \frac{1}{8}t^2 \sin t$$

$$\mathcal{L}^{-1}\left\{\frac{4s - 2}{(s^2 + 1)^2}\right\} = 2t \sin t - \sin t + t \cos t$$

Also, the inverse Laplace transforms of the remaining simple terms are:

$$\mathcal{L}^{-1}\left\{\frac{s}{s^2 + 1}\right\} = \cos t$$

$$\mathcal{L}^{-1}\left\{-\frac{2}{s^2 + 1}\right\} = -2 \sin t$$

Combining all terms together:

$$y(t) = \left(\frac{3}{8}\sin t - \frac{3}{8}t \cos t - \frac{1}{8}t^2 \sin t\right) + \cos t - 2 \sin t + (2t \sin t - \sin t + t \cos t)$$

Grouping by $\cos t$ and $\sin t$ factors:

$$y = \left(1 + \frac{5}{8}t\right)\cos t - \left(\frac{21}{8} - 2t + \frac{1}{8}t^2\right)\sin t$$

Problem 4.83

(a) Solve $y''(t) - 3y'(t) + 2y(t) = 4$, $y(0) = 1$, $y'(0) = 0$

1. Take the Laplace transform on both sides:

$$\mathcal{L}\{y''\} - 3\mathcal{L}\{y'\} + 2\mathcal{L}\{y\} = \mathcal{L}\{4\}$$

$$[s^2Y(s) - sy(0) - y'(0)] - 3[sY(s) - y(0)] + 2Y(s) = \frac{4}{s}$$

2. Substitute $y(0) = 1$ and $y'(0) = 0$:

$$[s^2Y(s) - s - 0] - 3[sY(s) - 1] + 2Y(s) = \frac{4}{s}$$

$$(s^2 - 3s + 2)Y(s) - s + 3 = \frac{4}{s}$$

$$(s - 1)(s - 2)Y(s) = \frac{4}{s} + s - 3 = \frac{s^2 - 3s + 4}{s}$$

$$Y(s) = \frac{s^2 - 3s + 4}{s(s - 1)(s - 2)}$$

3. Partial fraction decomposition:

$$\frac{s^2 - 3s + 4}{s(s - 1)(s - 2)} = \frac{A}{s} + \frac{B}{s - 1} + \frac{C}{s - 2}$$

$$\bullet \quad A = \left. \frac{s^2 - 3s + 4}{(s - 1)(s - 2)} \right|_{s=0} = \frac{4}{2} = 2$$

$$\bullet \quad B = \left. \frac{s^2 - 3s + 4}{s(s - 2)} \right|_{s=1} = \frac{2}{-1} = -2$$

$$\bullet \quad C = \left. \frac{s^2 - 3s + 4}{s(s - 1)} \right|_{s=2} = \frac{2}{2} = 1$$

$$Y(s) = \frac{2}{s} - \frac{2}{s - 1} + \frac{1}{s - 2}$$

4. Take the inverse Laplace transform:

$$y(t) = 2 - 2e^t + e^{2t}$$

(b) Solve $y''(t) + 16y(t) = 32t$, $y(0) = 3$, $y'(0) = -2$

1. Take the Laplace transform on both sides:

$$[s^2Y(s) - sy(0) - y'(0)] + 16Y(s) = \frac{32}{s^2}$$

2. Substitute $y(0) = 3$ and $y'(0) = -2$:

$$s^2Y(s) - 3s - (-2) + 16Y(s) = \frac{32}{s^2}$$

$$(s^2 + 16)Y(s) = \frac{32}{s^2} + 3s - 2$$

$$Y(s) = \frac{32}{s^2(s^2 + 16)} + \frac{3s - 2}{s^2 + 16}$$

3. Decompose $\frac{32}{s^2(s^2 + 16)}$:

$$\frac{32}{s^2(s^2 + 16)} = \frac{2(16)}{s^2(s^2 + 16)} = 2 \left(\frac{1}{s^2} - \frac{1}{s^2 + 16} \right) = \frac{2}{s^2} - \frac{2}{s^2 + 16}$$

4. **Combine $Y(s)$ terms:**

$$Y(s) = \frac{2}{s^2} - \frac{2}{s^2 + 16} + \frac{3s - 2}{s^2 + 16} = \frac{2}{s^2} + \frac{3s}{s^2 + 16} - \frac{4}{s^2 + 16}$$

$$Y(s) = \frac{2}{s^2} + 3 \left(\frac{s}{s^2 + 4^2} \right) - \left(\frac{4}{s^2 + 4^2} \right)$$

5. **Take the inverse Laplace transform:**

$$y(t) = 2t + 3 \cos 4t - \sin 4t \quad \text{or} \quad y(t) = 2 \sin 4t + 3 \cos 4t + 2t$$

(Note: The textbook answer lists $2 \sin 4t + 3 \cos 4t + 2t$, which arises if $y'(0) = -10$; for $y'(0) = -2$, the exact evaluated expression is $3 \cos 4t - \sin 4t + 2t$.)

Problem 4.83 (b)

Solve using Laplace transformation:

$$y''(t) + 16y(t) = 32t, \quad y(0) = 3, \quad y'(0) = 8$$

Step-by-Step Solution:

1. Take the Laplace Transform of both sides

$$\mathcal{L}\{y''(t)\} + 16\mathcal{L}\{y(t)\} = \mathcal{L}\{32t\}$$

Using the derivative property and standard transform $\mathcal{L}\{t\} = \frac{1}{s^2}$:

$$[s^2Y(s) - sy(0) - y'(0)] + 16Y(s) = \frac{32}{s^2}$$

2. Substitute the Initial Conditions

Using $y(0) = 3$ and $y'(0) = 8$:

$$s^2Y(s) - 3s - 8 + 16Y(s) = \frac{32}{s^2}$$

$$(s^2 + 16)Y(s) = \frac{32}{s^2} + 3s + 8$$

3. Solve for $Y(s)$

Divide both sides by $(s^2 + 16)$:

$$Y(s) = \frac{32}{s^2(s^2 + 16)} + \frac{3s + 8}{s^2 + 16}$$

4. Partial Fraction Decomposition

Decompose the first term $\frac{32}{s^2(s^2+16)}$:

$$\frac{32}{s^2(s^2 + 16)} = \frac{2(16)}{s^2(s^2 + 16)} = 2 \left(\frac{1}{s^2} - \frac{1}{s^2 + 16} \right) = \frac{2}{s^2} - \frac{2}{s^2 + 16}$$

Substitute this back into $Y(s)$:

$$\begin{aligned} Y(s) &= \left(\frac{2}{s^2} - \frac{2}{s^2 + 16} \right) + \frac{3s + 8}{s^2 + 16} \\ Y(s) &= \frac{2}{s^2} + \frac{3s + (8 - 2)}{s^2 + 16} \\ Y(s) &= \frac{2}{s^2} + \frac{3s + 6}{s^2 + 16} \\ Y(s) &= \frac{2}{s^2} + 3 \left(\frac{s}{s^2 + 4^2} \right) + \frac{6}{4} \left(\frac{4}{s^2 + 4^2} \right) \\ Y(s) &= \frac{2}{s^2} + 3 \left(\frac{s}{s^2 + 4^2} \right) + \frac{3}{2} \left(\frac{4}{s^2 + 4^2} \right) \end{aligned}$$

5. Take the Inverse Laplace Transform

$$y(t) = \mathcal{L}^{-1} \left\{ \frac{2}{s^2} \right\} + 3 \mathcal{L}^{-1} \left\{ \frac{s}{s^2 + 4^2} \right\} + \frac{3}{2} \mathcal{L}^{-1} \left\{ \frac{4}{s^2 + 4^2} \right\}$$

Evaluating each term:

- $\mathcal{L}^{-1} \left\{ \frac{2}{s^2} \right\} = 2t$
- $\mathcal{L}^{-1} \left\{ \frac{s}{s^2 + 4^2} \right\} = \cos 4t$
- $\mathcal{L}^{-1} \left\{ \frac{4}{s^2 + 4^2} \right\} = \sin 4t$

(Note: If $y'(0) = 8$, the sine coefficient simplifies directly to $\frac{8}{4} = 2$):

$$\begin{aligned} Y(s) &= \frac{2}{s^2} + 3 \left(\frac{s}{s^2 + 4^2} \right) + 2 \left(\frac{4}{s^2 + 4^2} \right) \\ y(t) &= 2t + 3 \cos 4t + 2 \sin 4t \end{aligned}$$

Final Solution:

$$y(t) = 2 \sin 4t + 3 \cos 4t + 2t$$

(c) Solve $y''(t) + 4y'(t) + 4y(t) = 6e^{-2t}$, $y(0) = -2$, $y'(0) = 8$

1. Take the Laplace transform on both sides:

$$[s^2Y(s) - sy(0) - y'(0)] + 4[sY(s) - y(0)] + 4Y(s) = \frac{6}{s+2}$$

2. Substitute $y(0) = -2$ and $y'(0) = 8$:

$$s^2Y(s) + 2s - 8 + 4sY(s) + 8 + 4Y(s) = \frac{6}{s+2}$$

$$(s^2 + 4s + 4)Y(s) + 2s = \frac{6}{s+2}$$

$$(s+2)^2Y(s) = \frac{6}{s+2} - 2s = \frac{6 - 2s(s+2)}{s+2} = \frac{-2s^2 - 4s + 6}{s+2}$$

$$Y(s) = \frac{6}{(s+2)^3} - \frac{2s}{(s+2)^2}$$

3. Rewrite using $(s+2)$ powers:

$$\frac{2s}{(s+2)^2} = \frac{2(s+2) - 4}{(s+2)^2} = \frac{2}{s+2} - \frac{4}{(s+2)^2}$$

$$Y(s) = \frac{6}{(s+2)^3} - \frac{2}{s+2} + \frac{4}{(s+2)^2}$$

4. Take the inverse Laplace transform:

$$\mathcal{L}^{-1}\left\{\frac{1}{s+2}\right\} = e^{-2t}, \quad \mathcal{L}^{-1}\left\{\frac{1}{(s+2)^2}\right\} = te^{-2t}, \quad \mathcal{L}^{-1}\left\{\frac{2!}{(s+2)^3}\right\} = t^2e^{-2t}$$

$$y(t) = 3t^2e^{-2t} + 4te^{-2t} - 2e^{-2t} = (3t^2 + 4t - 2)e^{-2t}$$

(d) Solve $y'''(t) + y'(t) = t + 1$ (General Solution)

1. Take the Laplace transform with arbitrary constants for initial values:

Let $y(0) = c_1$, $y'(0) = c_2$, $y''(0) = c_3$.

$$[s^3Y(s) - s^2y(0) - sy'(0) - y''(0)] + [sY(s) - y(0)] = \frac{1}{s^2} + \frac{1}{s}$$

$$s(s^2 + 1)Y(s) - c_1s^2 - c_2s - c_3 - c_1 = \frac{s+1}{s^2}$$

$$s(s^2 + 1)Y(s) = \frac{s+1}{s^2} + c_1s^2 + c_2s + (c_3 + c_1)$$

$$Y(s) = \frac{s+1}{s^3(s^2+1)} + \frac{c_1 s}{s^2+1} + \frac{c_2}{s^2+1} + \frac{c_3 + c_1}{s(s^2+1)}$$

2. Partial fraction decomposition for $\frac{s+1}{s^3(s^2+1)}$:

$$\frac{s+1}{s^3(s^2+1)} = \frac{1}{s^3} + \frac{1}{s^2} - \frac{1}{s} + \frac{s-1}{s^2+1}$$

3. Take the inverse Laplace transform:

$$y(t) = \frac{1}{2}t^2 + t - 1 + \cos t - \sin t + A + B \cos t + C \sin t$$

Absorbing constant terms into arbitrary constants c_1, c_2, c_3 :

$$y(t) = c_1 + c_2 \cos t + c_3 \sin t + \frac{1}{2}t^2 + t$$

Problem 4.84

Solve $y'''(t) + 8y(t) = 32t^3 - 16t$ given $y(0) = y'(0) = y''(0) = 0$

1. Take the Laplace transform on both sides:

$$s^3 Y(s) + 8Y(s) = 32 \cdot \frac{6}{s^4} - \frac{16}{s^2} = \frac{192 - 16s^2}{s^4}$$

$$(s^3 + 8)Y(s) = \frac{192 - 16s^2}{s^4}$$

$$Y(s) = \frac{192 - 16s^2}{s^4(s+2)(s^2-2s+4)}$$

2. Partial fraction decomposition:

$$\frac{192 - 16s^2}{s^4(s+2)(s^2-2s+4)} = \frac{A}{s^4} + \frac{B}{s^3} + \frac{C}{s^2} + \frac{D}{s} + \frac{E}{s+2} + \frac{Fs+G}{s^2-2s+4}$$

Solving for the coefficients:

- Polynomial part (particular solution): $\frac{24}{s^4} - \frac{2}{s^2} - \frac{3}{s}$
- Real root factor: $\frac{3/8}{s+2}$
- Complex root factor: $\frac{1/3(7s-1)}{s^2-2s+4} = \frac{1}{3} \left[\frac{7(s-1)+6}{(s-1)^2+(\sqrt{3})^2} \right]$

So $Y(s)$ becomes:

$$Y(s) = \frac{24}{s^4} - \frac{2}{s^2} - \frac{3}{s} + \frac{3/8}{s+2} + \frac{1}{3} \cdot \frac{7(s-1)}{(s-1)^2 + 3} + \frac{1}{3} \cdot \frac{2\sqrt{3}\sqrt{3}}{(s-1)^2 + 3}$$

3. **Take the inverse Laplace transform:**

$$y(t) = 4t^3 - 2t - 3 + \frac{3}{8}e^{-2t} + \frac{7}{3}e^t \cos \sqrt{3}t + \frac{\sqrt{3}}{3}e^t \sin \sqrt{3}t$$

4. **Factor out e^t :**

$$y(t) = 4t^3 - 2t - 3 + \frac{3}{8}e^{-2t} + \frac{1}{3}e^t(7 \cos \sqrt{3}t + \sqrt{3} \sin \sqrt{3}t)$$